Partial Gluing and Level Two: Graded Explanations, Latent Actors and Paths through Singularities

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Abstract

When local tables of evidence fail to glue into one global law, the usual response is to stop. We stay at the failure and ask how much structure can still be saved. For a family of tables with noncontextual fraction $\lambda^* < 1$, the laws that account for a fraction α of every table form a polytope E_α. It is non-empty exactly when $\alpha \le \lambda^*$, it shrinks as α grows, and its prediction intervals and its maximum-entropy member move continuously with α. The gradation α is how much of the gluing we keep; the rest, $1-\alpha$ of every table, is mass we let dissipate. We then go up one level, to laws over hypotheses: latent types of patients (actors) whose mixture produces the observed law. The explanations at this level are the real points of the fibre of a polynomial map, an affine scheme, and its geometry carries what happens to the actors. Where two types coincide the fibre is singular, a second-order obstruction κ (the first non-abelian term, $\ell_2 \neq 0$) is non-zero, and the tangent cone is a union of planes: types can split apart along one variable for free but not along two. Along a straight path of observed laws the types separate like $\sqrt{s}$ (two Puiseux branches); on the other side of the independence locus the two explanations become complex conjugates, no real two-type explanation exists, and a third type is needed. Going around a loop of populations can exchange the roles of the two actors ($\mathbb{Z}/2$ monodromy). With three measured variables this exchange must pass through a singularity; with four it need not. We illustrate everything on a drug-combination problem: baseline and single-drug harms are known, pairs have been trialled, the triple combination has not. Latent types narrow the prediction interval about fivefold; when the pair trials conflict, the graded explanations say what can be predicted when 20%, 40%, 70% or 86% of the structure is saved. Finally we make the scheme structure explicit. In mean-and-difference coordinates the model is a toric family over the weights; the fibre over an independent law is exactly the Stanley–Reisner scheme of a small simplicial complex whose faces are the strata (it is reduced); and for three variables the model inverts in closed form, flatness fails exactly where an actor is absent or the actors agree on a variable, and the sign of Cayley's hyperdeterminant separates genuine mixtures, signed mixtures and complex-conjugate pairs. Real explanations can vanish only at a wall of the parameter box or at the non-flat locus, and this is why role exchange needs a singularity with three variables but not with four. Both levels then sit on one ladder of base semirings, $\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\ge 0}$: disagreement on an overlap has no points at all; contextuality has real points but no positive ones, with negativity comparable to $1-\lambda^*$; and at level two a further rung of complex-conjugate actors appears. Finally we let the coefficients be matrices and keep what fails to commute: commutators decide the cover of contexts (CHSH gives $\lambda^* = 2-\sqrt{2}$), order effects are bounded by commutator norms, the canonical Kirkwood–Dirac section puts the commutation obstruction on the complex rung, loops carry a geometric phase that becomes non-abelian with rank-two projections, and non-commuting projections steer a state along a probability

path (Zeno dragging) where commuting ones cannot. With the measurements given but partly masked, the linear fit locates a hidden direction exactly when the known directions on its side span space, leaving otherwise a linear ambiguity of computable dimension; positivity then culls the candidates that would imply negative probabilities, and what remains recoverable are the invariants of the residual symmetry. Last, quantum graded gluing is the positivity domain of a hereditary quadratic matrix polynomial in the sense of Islam, Volčič and Woerdeman: its walls are determinantal free loci, and the quantum fraction lies below the classical fraction of every product-basis view, with equality for classical data and, with a proof by symmetry, for pairwise Werner windows, where it exhibits monogamy. A final section gives these pieces a common categorical home: the contexts are a diagram of commutative subalgebras and contextuality a failed limit over it (Kochen–Specker versus Gleason, exact on the Peres–Mermin square); representation functors explain the gauge and show that a holonomy phase q allows finite dimension only when $q^n = 1$; positivity on truncated free modules (NPA) gives the quantum rung without fixing a dimension, with closed forms on isotropic boxes; and divergence-enriched categories give a data processing inequality for λ^*. All claims are checked: nine numerical checks (`engine/check_level2.py`) six exact symbolic ones (`engine/check_scheme.py`), six on the ladder (`engine/check_ladder.py`) eight on non-commutativity (`engine/check_noncomm.py`) seven on masked measurements (`engine/check_masked.py`, `engine/check_masked_gauge.py`) seven on quantum graded gluing (`engine/check_qglue.py`) and six on the categorical home (`engine/check_npa.py`).

1 The problem: a combination nobody has trialled

Three drugs A, B, C are each known by themselves: single-drug trials give the rate of harm under each. Pairs have been trialled: from the AB, BC and AC trials we have the joint tables of harm. The triple combination has not been given to anyone. Write $H_A, H_B, H_C \in \{0, 1\}$ for whether a patient is susceptible to the harm of each drug, and suppose for the illustration that a patient on a combination is harmed when susceptible to at least one of its drugs. The question is

$$P(\text{any harm} \mid A{+}B{+}C) \;=\; 1 - P(H_A = H_B = H_C = 0),$$

a property of the joint law of (H_A, H_B, H_C), which no trial has observed. The baseline (no drug, no harm) and the single-drug margins are the known blobs; the joint law is the unknown blob; the pair tables are *windows* onto it.

Three things can happen.

1. The pair tables glue, and the set of global laws consistent with them gives an interval for the triple harm (level one).
2. Patients come in latent types, actors that are invisible in the data but shape it: a type susceptible to A and C, say, and a type susceptible to B. The same patient can play different roles in different pairs. If we are willing to assume a small number of types, the interval narrows (level two, section 3).
3. The pair tables do not glue: the BC trial was run in a different population. The standard answer is that there is no global law and nothing to predict. We stay at the failure and grade it (section 2).

The rest of the note develops each, and then follows the latent actors through the singular places of their geometry: where they merge, where one of them disappears, where a new one must be born, and where they exchange roles.

Reading guide. The note uses two towers: the posterior tower of [82, 83] (depth: laws, laws of laws, shadows) and the ladder of base semirings $\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\geq 0}$ (coefficients). They are two axes of one grid, and every result has an address in it; section 16 and table 9 give the map and may be read first.

2 Sitting at the gluing failure: graded explanations

Let $X = (X_v)_{v \in V}$ take values in finite sets, and let a family of windows w have scopes $S_w \subseteq V$ and tables t_w, laws on the values of X_{S_w}. For a law q on the values of X write $q|_w$ for its marginal on S_w. The *noncontextual fraction* [3] is

$$\lambda^* \;=\; \max\{\, \mu(\text{all}) : \mu \geq 0 \text{ a measure on the values of } X,\ \mu|_w \leq t_w \text{ for every } w \,\},$$

a linear program. The tables glue (there is a global law with these marginals) iff $\lambda^* = 1$.

Definition 1 (graded explanation)**.** For $\alpha \in (0, 1]$, an *α-explanation* is a law q with $\alpha\, q|_w \leq t_w$ for every window. Write E_α for the set of them. For an event F, the *α-prediction interval* is $[\min_{q \in E_\alpha} q(F),\ \max_{q \in E_\alpha} q(F)]$ (two linear programs), and the *most probable explanation* q_α is the maximiser of the entropy $H(q)$ over E_α.

An α-explanation splits every table in two,

$$t_w \;=\; \alpha\, q|_w \;+\; (1-\alpha)\, r_w, \qquad r_w = \frac{t_w - \alpha q|_w}{1-\alpha} \ \text{ a law on } S_w, \tag{1}$$

a part $\alpha q|_w$ that comes from one global law, and a residue r_w that comes from no common source: the residues of different windows need not agree on their overlaps. The fraction α is the structure we save; $1 - \alpha$ of the mass of every table is let go. This is the gradation in gluing.

Proposition 2. *(i) $E_\alpha \neq \emptyset$ iff $\alpha \leq \lambda^*$.*

(ii) If $\alpha \leq \beta$ then $E_\beta \subseteq E_\alpha$. Hence prediction intervals are nested: saving more structure never widens them.

(iii) On $(0, \lambda^*]$ the map $\alpha \mapsto E_\alpha$ is continuous (in the Hausdorff metric), and so are the endpoints of every prediction interval and the most probable explanation q_α, which is unique.

(iv) If all table entries are positive, E_α is the whole simplex for small α: the interval is $[0, 1]$ for any event that is neither sure nor impossible, and q_α is the uniform law.

Proof. (i) If $q \in E_\alpha$ then $\mu = \alpha q$ is feasible for the program defining λ^* with total mass α, so $\alpha \leq \lambda^*$. Conversely, if μ attains λ^* and $\alpha \leq \lambda^*$, then $q = \mu/\lambda^*$ satisfies $\alpha q|_w \leq \lambda^* q|_w = \mu|_w \leq t_w$. (ii) $\alpha q|_w \leq \beta q|_w \leq t_w$. (iii) Let $M = \{\mu \geq 0 : \mu|_w \leq t_w\}$, a convex polytope; $E_\alpha = \{\mu/\alpha : \mu \in M,\ \mu(\text{all}) = \alpha\}$. The correspondence $\alpha \mapsto E_\alpha$ has a closed graph and compact values, so it is upper hemicontinuous. It is also lower hemicontinuous: given $q \in E_\alpha$ and $\alpha_n \to \alpha$, if $\alpha_n \leq \alpha$ then $q \in E_{\alpha_n}$ by (ii); if $\alpha < \alpha_n \leq \lambda^*$, take $\mu^* \in M$ of mass λ^* and $s_n \to 0$ with $(1 - s_n)\alpha + s_n \lambda^* = \alpha_n$; then $\mu_n = (1 - s_n)\alpha q + s_n \mu^* \in M$ (by convexity) has mass α_n, and $q_n = \mu_n/\alpha_n \in E_{\alpha_n}$ tends to q. A continuous compact-valued correspondence has continuous maxima and minima of a continuous function (Berge's maximum theorem [5]); the entropy is strictly concave, so its maximiser is unique and continuous. (iv) If every entry of every table is positive, every law q satisfies $\alpha q|_w \leq t_w$ once $\alpha \leq \min t_w$, so E_α is the whole simplex for small α; zero entries in the tables force the corresponding cells to zero for every α, and the statement holds on the remaining face. □

So the graded explanations form a path. At $\alpha \to 0$ we have saved nothing and know nothing; at $\alpha = \lambda^*$ we have saved all the structure that can be saved together. In between, $\alpha \mapsto q_\alpha$ is the *most probable path* from ignorance to the best partial gluing, and the prediction band around it narrows monotonically.

Example 3 (conflicting pair trials)**.** Let the truth in population 1 be two patient types with weights $(0.6, 0.4)$ and susceptibilities

$$a_1 = (0.5,\ 0.1,\ 0.4), \qquad a_2 = (0.05,\ 0.5,\ 0.1) \qquad (\text{for } H_A, H_B, H_C),$$

the harms independent within a type. The AB and AC trials sample population 1; the BC trial samples population 2, with the same types in proportions $(0.25, 0.75)$. Each table is exact, but together they do not glue: $\lambda^* = 0.86$. The triple-combination harm is 0.667 in population 1 and 0.612 in population 2. Table 1 and fig. 1(a) show the graded answer.

structure saved α	prediction interval	most probable q_α	reading
0.2	[0, 1]	0.867	nothing ruled out; close to the uniform law (7/8)
0.4	[0, 1]	0.835	
0.6	[0.233, 0.976]	0.781	the trials begin to bite
0.7	[0.343, 0.836]	0.765	
0.8	[0.488, 0.732]	0.694	
$0.86 = \lambda^*$	[0.594, 0.681]	0.647	all that can be saved; both populations' truths inside
0.87	none	none	cannot be saved together

Table 1: Graded partial gluing of the conflicting pair trials (example 3): the triple-combination harm when a fraction α of every table is explained by one global law and $1 - \alpha$ is let go.

At $\alpha = \lambda^*$ the interval $[0.594, 0.681]$ contains the triple harm of both populations. This is not a theorem (it is a property of this example), but it shows what the graded answer is for: when the evidence conflicts, the interval at the largest savable fraction is an honest statement of what is still predictable, and the residues r_w in (1) say where the conflict lives.

3 Level two: latent actors

Level one asks for a law on outcomes. Level two asks for a law on hypotheses that produces it: here, a mixture of patient types.

Definition 4 (latent class model)**.** With k types and m binary variables, type h has weight w_h and susceptibilities $a_h \in [0,1]^m$, the variables independent given the type. The observed law is

$$F(w, a) \ = \ \sum_{h=1}^{k} w_h \, \Pi(a_h), \qquad \Pi(a_h) = \mathrm{Bern}(a_{h1}) \otimes \cdots \otimes \mathrm{Bern}(a_{hm}),$$

a polynomial map from $\Delta_{k-1} \times [0,1]^{km}$ to the simplex of laws on $\{0,1\}^m$.

For an observed law P, the level-two explanations are the real points, with $w \geq 0$ and $a \in [0,1]^{km}$, of

$$\mathcal{X}_P \ = \ \mathrm{Spec}\ \mathbb{R}[w, a] \big/ \big(\textstyle\sum_h w_h - 1,\ F(w, a) - P \big),$$

the fibre of F over P, an affine scheme. When only windows are observed, F is composed with the marginal maps, and the fibre is taken over the observed tables. Level one is the image; level two is the fibre. Everything that happens to the actors happens in the fibre: a fibre that is a finite set of points (identification), a curve (a one-parameter family of explanations), a point where two branches cross (actors that merge), or empty (no explanation with this many actors).

For $k = 2$ and $m = 3$ there are $1 + 2m = 7$ parameters and the laws on $\{0,1\}^3$ form a 7-dimensional simplex; generically the fibre over the full law is two points, exchanged by swapping the labels of the types [4]. Over the three pair tables (6 numbers after normalisation) the fibre is a curve.

Example 5 (latent structure sharpens prediction)**.** In population 1 of example 3, with all three pair trials run there, the pair tables glue. Level one (any global law with these pair tables) gives the triple harm in [0.654, 0.694]. Level two (any mixture of two types with these pair tables) gives [0.6613, 0.6687], about five times narrower, and still containing the truth 0.667 (fig. 1(b)). The level-two interval was computed by tracing the fibre curve: we fix w_1 on a grid of 49 values and solve for the susceptibilities from 12 starts each; 228 exact solutions were found.

Assuming latent actors is a modelling choice and the narrower interval is only as good as that choice. What the fibre adds is that the choice is explicit and its consequences can be computed, including where it breaks.

Graded gluing at level two. The gradation of section 2 lifts: the level-two gluable fraction is

$$\lambda_2^* \;=\; \sup\{\alpha :\; \alpha\, F(w,a)|_w \le t_w \text{ for every window, for some } (w,a)\},$$

the largest share of every table that a k-type population can account for. Since $F(w,a)$ is a law, $\lambda_2^* \le \lambda^*$. For the conflicting trials of example 3 the best two-type explanation found saves 0.86, the same as level one: there, assuming two actors costs no structure. Unlike λ^*, λ_2^* is not a linear program (it is a polynomial optimisation), and the values we report are the best found from many starts, so they are lower bounds.

4 Singularities and the second-order obstruction

Let $J = dF$ at a point $x = (w,a)$, restricted to directions with $\sum_h \delta w_h = 0$, and viewed as a map into the sum-zero directions of the simplex. Its kernel K is the space of first-order deformations of the explanation that leave the observed law unchanged; let C be a basis of its cokernel. Define the *second-order obstruction*

$$\kappa(v) \;=\; C^\top D^2F(v,v), \qquad v \in K.$$

Proposition 6 (Kuranishi test [19])**.** *If $\gamma(t)$ is a smooth path in the fibre with $\gamma(0) = x$ and $\gamma'(0) = v$, then $v \in K$ and $\kappa(v) = 0$.*

Proof. Differentiate $F(\gamma(t)) = P$ twice at $t = 0$: $Jv = 0$ and $J\gamma''(0) + D^2F(v,v) = 0$, so $D^2F(v,v) \in \operatorname{im} J$, which $C^\top$ annihilates. □

At a regular point $K = 0$ and nothing is obstructed. The obstruction is the first term of the deformation problem that is not linear: in the language of L_∞ algebras [20] it is the bracket ℓ_2, and at level one (where the explanation sets are polytopes) it vanishes identically. Level two is where it first appears. The fibre has three kinds of points (table 2).

point	meaning	rank J	dim K	obstruction
generic	two distinct, present actors	$2m+1=7$	0	none
$w_2=0$	an actor absent	$m+1=4$	$m=3$	$\kappa=0$: its role is free
$a_1=a_2$	two actors coincide	$m=3$	$m+1=4$	$\kappa\neq 0$ off a union of planes

Table 2: Strata of the two-type, three-variable fibre ($m=3$). An absent actor can take any role at no cost; coinciding actors can be pulled apart only along special directions.

Theorem 7 (tangent cone at a coincidence)**.** *Let $a_1=a_2=c\in(0,1)^m$ and $w\in(0,1)^2$. The kernel K is spanned by the weight direction $(\delta w,0)$ with $\delta w=(1,-1)$ and the* splits $s(e)=(0;\ -\frac{w_2}{w_1}e,\ e)$, $e\in\mathbb{R}^m$. *For $v=t\,(\delta w,0)+s(e)$,*

$$\kappa(v)\ =\ \frac{2w_2}{w_1}\sum_{i<j}e_ie_j\ C^\top\partial_{ij}\Pi(c),$$

which vanishes iff e has at most one non-zero coordinate. The tangent cone of the fibre at the coincidence is therefore the union of the m planes $\langle(\delta w,0),\ s(\mathbf{e}_i)\rangle$, which meet along the weight line, and each plane is filled by exact paths in the fibre.

Proof. Write $\partial_S\Pi(c)$ for Π with the factors $i\in S$ replaced by $(-1,1)$. The 2^m vectors $\partial_S\Pi(c)$ form a basis (a tensor product of bases $\{\mathrm{Bern}(c_i),(-1,1)\}$), and those with $S\neq\emptyset$ span the sum-zero directions. The columns of J are $\Pi(a_1)-\Pi(a_2)=0$ (weights) and $w_h\partial_i\Pi(c)$ (susceptibilities), so $\operatorname{im}J=\langle\partial_i\Pi\rangle$, the kernel is as stated, and the cokernel is spanned by $\partial_S\Pi$ with $|S|\geq 2$. In $D^2F(v,v)=2\sum_h\sum_i\delta w_h\delta a_{hi}\partial_i\Pi+2\sum_h w_h\sum_{i<j}\delta a_{hi}\delta a_{hj}\partial_{ij}\Pi$ (there are no ∂_{ii} terms, F being multilinear in each a_h), the first sum lies in $\operatorname{im}J$ and the second equals $2\big(w_1\frac{w_2^2}{w_1^2}+w_2\big)\sum_{i<j}e_ie_j\partial_{ij}\Pi=\frac{2w_2}{w_1}\sum_{i<j}e_ie_j\partial_{ij}\Pi$. Since the $\partial_{ij}\Pi$ are independent modulo $\operatorname{im}J$, $\kappa(v)=0$ iff $e_ie_j=0$ for all $i<j$. For the paths: if two types differ only in variable i, the mixture is still a product law, whose i-th margin is $w_1a_{1i}+w_2a_{2i}$; moving the weights and spreading a_{1i},a_{2i} while keeping this mean fixed stays in the fibre exactly. □

A mixture of two products that differ in one variable is itself a product. Actors that differ in only one respect are therefore invisible: the fibre contains a whole plane of them through the coincidence. Actors that differ in two respects or more create a correlation, and they cannot be separated infinitesimally without changing the observed law at second order. Figure 1(d) shows $|\kappa|$ around the circle of splits in the (X_1,X_2) plane: it vanishes exactly on the two coordinate axes and is proportional to $|\sin 2\varphi|$ in between.

5 Where mass dissipates: paths through singularities

Now move the observed law. Fix $w=(\frac12,\frac12)$, $c\in(0,1)^3$ and $e\in\mathbb{R}^3$ with all $e_i\neq 0$, and set $B=\sum_{i<j}e_ie_j\,\partial_{ij}\Pi(c)$ and $P(s)=\Pi(c)+sB$, a straight line of laws through the product law $\Pi(c)$.

Theorem 8 (two branches, then none)**.** *(i) For $s>0$ (small enough that the parameters stay in $[0,1]$), $P(s)=F\big(w,\ (c+\sqrt{s}\,e,\ c-\sqrt{s}\,e)\big)$ exactly. The two types separate like $\sqrt{s}$, and the two Puiseux branches $\pm\sqrt{s}$ [13] are exchanged by swapping labels.*

(ii) For $s<0$ no mixture of two (or one) product laws equals $P(s)$.

(iii) At $s=0$ the path passes through the coincidence stratum of theorem 7.

Proof. (i) $\Pi(c+te) = \sum_S t^{|S|} e_S\, \partial_S \Pi(c)$ with $e_S = \prod_{i\in S} e_i$. The average of t and $-t$ keeps the even terms: $\Pi(c) + t^2 B$ for $m = 3$. Put $t^2 = s$. (ii) Under $P(s)$ the covariance of X_i, X_j is $s\, e_i e_j$, so the product of the three pairwise covariances is $s^3 (e_1 e_2 e_3)^2 < 0$. Under any two-type law it is $\prod_{i<j} w_1 w_2 d_i d_j = (w_1 w_2)^3 (d_1 d_2 d_3)^2 \geq 0$, where $d = a_1 - a_2$. □

This is the precise form of an abrupt change. The observed law moves linearly in s; the explanation moves like $\sqrt{s}$, with infinite speed at $s = 0$; and on the far side the two-type structure is gone from the real world: the same formula gives $c \pm i\sqrt{|s|}\, e$, a complex-conjugate pair of explanations (section 7 makes this exact). In the space of observed laws the image of F has a fold along the independence locus, and the path crosses it. Figure 1(c) shows the gap $\max_i |a_{1i} - a_{2i}| = 2\sqrt{s}\max_i |e_i|$ fitted from the data, exactly on the curve, and the best two-type misfit for $s < 0$: 0.0004, 0.0008, 0.0017 and 0.0035 at $s = -0.005, -0.01, -0.02, -0.04$.

Graded gluing says how much is lost. Beyond the fold $\lambda_2^* < 1$: some mass of every observed law must be let go if two actors are to explain the rest. The best two-type explanations we found save 0.9990, 0.9926, 0.9960, 0.9919 and 0.9836 of the structure at $s = -0.0025$, -0.005, -0.01, -0.02, -0.04: about 1% of the mass must go, a little more as s moves away from the fold. These are attained values and so lower bounds on λ_2^*; that they are not monotone in s shows the optimiser does not resolve the exact rate. The mass does not vanish into nothing, however: three types fit $P(s)$ exactly for $s = -0.01$ and $s = -0.04$ (residual below 10^{-14}). What dissipates is the two-actor organisation of the mass, and what is born on the far side is a third actor. Read in the other direction, the fold is where a type dies: as $s \downarrow 0$ the two types merge, and their difference, which had carried a correlation, becomes a free direction of the kind in table 2.

6 Dual roles and monodromy

The labels "type 1" and "type 2" are not in the data. Over a generic law the fibre is two points that differ only in the labels, and along a path of laws the labels can be carried by continuity. Around a closed loop of laws they may come back exchanged: the actor that played the A-susceptible role now plays the B-susceptible one. This is the monodromy of the map F, here a subgroup of $\mathbb{Z}/2$ (of the symmetric group S_k for k types).

We build such loops explicitly. With a centre c, directions $u, v \in \mathbb{R}^m$, $r = 0.22$ and $\rho = 0.15$, put

$$a_{1,2}(\theta) = c \pm r(\cos\theta\, u + \sin\theta\, v), \qquad w(\theta) = \big(\tfrac{1}{2} + \rho\cos\theta,\ \tfrac{1}{2} - \rho\cos\theta\big), \qquad \theta \in [0, \pi].$$

At $\theta = \pi$ the parameters are those at $\theta = 0$ with the labels swapped, so the observed law $F(\theta)$ returns to its start (to 10^{-12}): a half circle of parameters is a closed loop of observed laws, and tracking the explanation along it exchanges the actors.

Theorem 9 (three variables force a singularity)**.** *Let $m = 3$, $k = 2$. Any continuous path of parameters with $w \in (0,1)^2$ from (w, a_1, a_2), with all $d_i = a_{1i} - a_{2i} \neq 0$, to its relabelling (w_2, w_1, a_2, a_1) passes, for each variable i, through a point where $d_i = 0$, and there* $\operatorname{rank} J \leq 5 < 7$. *So within the regular locus no role exchange is possible.*

Proof. Along the path d_i goes from d_i to $-d_i$ and so vanishes somewhere. Where $d_3 = 0$, say, the law is $G(w, a_{\cdot,1:2}) \otimes \mathrm{Bern}(c_3)$, where G is the two-type model on the first two variables. G has 5 parameters and a 3-dimensional target, so its differential has a kernel of dimension at least 2; those directions (with $\delta a_{\cdot 3} = 0$) are in the kernel of J. □

With four variables the same argument leaves a two-type model on three variables, which is generically regular, and nothing forces a singularity. Our 4-variable loop ($u = (1, 0.6, -0.4, 0.2)$, $v = (0.3, 1, 0.8, -0.9)$, $c = \frac{1}{2}\mathbf{1}$) keeps the smallest singular value of J at or above 0.0072 along 500 points of the half loop, the tracked solution ends at the swapped labels, and the full loop returns the original labels: genuine $\mathbb{Z}/2$ monodromy in the regular locus (fig. 1(e,f)). This is numerical evidence for the loop, not a proof. The 3-variable loop ($u = (1, 0.6, -0.4)$, $v = (0.3, 1, 0.8)$) shows the three forced dips of theorem 9, one for each variable, where the smallest singular value falls to the order of 10^{-6}.

In the drug setting: with three measured harms, two patient types cannot exchange their roles without passing through a population in which they look alike in some harm; with four harms they can. Which actor is "the one susceptible to A" is a local name, defined along a path, not a global fact; the observed law, at level one, is single-valued and never notices.

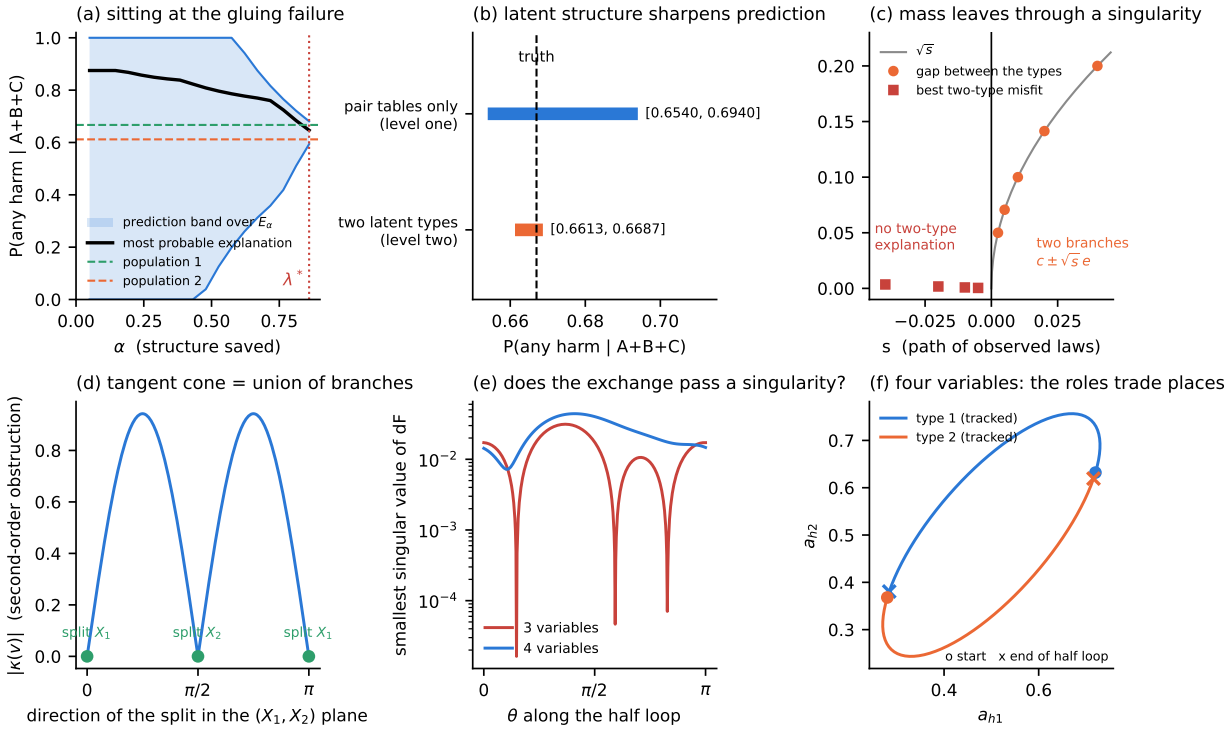


Figure 1: (a) Graded explanations for the conflicting trials: the prediction band over E_α, the most probable explanation, and the two populations' truths. (b) Level one against two latent types (population 1). (c) Crossing the independence locus: the gap between the types follows $\sqrt{s}$; for $s < 0$ no two-type explanation fits. (d) The second-order obstruction around the circle of splits at a coincidence: zero exactly on the coordinate branches. (e) Smallest singular value of dF along a role-exchanging half loop: with three variables it dips to zero three times, with four it stays away from zero. (f) The 4-variable half loop: the tracked types trade places.

7 Scheme structure: the fibre over independence and the flattening stratification

So far "scheme" has named the equations of the fibre. This section uses the scheme structure: ideals, their decompositions, reducedness and flatness. All statements here are checked exactly by Gröbner-basis computations [15] in SymPy [29] (`engine/check_scheme.py`, SC1–SC6).

Coordinates. Replace the susceptibilities by the mean and the difference of the two types,

$$\mu = w_1 a_1 + w_2 a_2, \qquad d = a_1 - a_2, \qquad \text{so} \qquad a_1 = \mu + w_2 d, \quad a_2 = \mu - w_1 d,$$

a polynomial automorphism of the parameter space (given w). On the observed side use the mean μ and the central moments $m_S = E\prod_{i\in S}(X_i - \mu_i)$, $|S| \geq 2$: the law and (μ, m) determine each other polynomially (unitriangularly).

Proposition 10 (moment normal form; SC1)**.** *$m_S = c_{|S|}(w)\, d_S$ with $d_S = \prod_{i\in S} d_i$ and $c_k(w) = w_1 w_2^k + w_2(-w_1)^k = w_1 w_2\big(w_2^{k-1} - (-w_1)^{k-1}\big)$.*

Proof. $a_{1i} - \mu_i = w_2 d_i$ and $a_{2i} - \mu_i = -w_1 d_i$, so $m_S = w_1(w_2)^{|S|} d_S + w_2(-w_1)^{|S|} d_S$. □

For fixed weights the map $d \mapsto (c_{|S|} d_S)_S$ is monomial: the two-type model is a toric family over the line of weights. These are the *secant cumulants* of Michałek, Oeding and Zwiernik [24], in which the secant variety becomes toric. Note $c_2 = w_1 w_2$ and $c_3 = w_1 w_2 (w_2 - w_1)$: an evenly mixed pair leaves no three-way moment.

The fibre over an independent law. Let $P = \Pi(c)$ be a product law (the variables independent). Its fibre is where the actors have merged, or one has gone.

Theorem 11 (the fibre is a Stanley–Reisner scheme; SC2)**.** *The ideal of the fibre over $\Pi(c)$ is*

$$I = (\mu - c) \;+\; \big(w_1 w_2\, d_i d_j :\; i < j\big) \;=\; (\mu - c) + \Big[(w_1) \cap (w_2) \cap \bigcap_{i=1}^{m} (d_j :\; j \neq i)\Big].$$

It is an intersection of prime ideals, hence radical: the fibre scheme is reduced. With $\mu = c$ it is the Stanley–Reisner scheme (cut by $w_1 + w_2 = 1$) of the simplicial complex Δ on the vertices $\{w_1, w_2, d_1, \ldots, d_m\}$ with facets

$$\{w_1, d_1, \ldots, d_m\}, \qquad \{w_2, d_1, \ldots, d_m\}, \qquad \{w_1, w_2, d_i\}\ (i = 1, \ldots, m).$$

Proof. By proposition 10 the fibre is $\mu = c$ and $c_{|S|}(w) d_S = 0$ for all $|S| \geq 2$. Every c_k is divisible by $c_2 = w_1 w_2$ and every d_S with $|S| \geq 3$ by some $d_i d_j$, so the ideal is $(\mu - c) + (w_1 w_2 d_i d_j)$. Since w_1 and $w_2 = 1 - w_1$ are coprime, $(w_1 w_2) = (w_1) \cap (w_2)$; the ideal $(d_i d_j : i < j)$ is the Stanley–Reisner ideal of m isolated points, $\bigcap_i (d_j : j \neq i)$; and the two involve disjoint variables, so their intersection is their product. The facets are the complements of the minimal non-faces $\{w_1, w_2, d_i, d_j\}$ in the sense of Stanley–Reisner theory [25]. The computation was also checked directly (Gröbner bases for $m = 3, 4$, and the intersection by elimination). □

The facets are the strata of table 2 named combinatorially: $\{w_1, d_1, \ldots, d_m\}$ is "actor 2 absent, actor 1's difference free", $\{w_2, \ldots\}$ the reverse, and $\{w_1, w_2, d_i\}$ "both present, differing only in variable i". Away from $w_1 w_2 = 0$ the factor $w_1 w_2$ is a unit, so near an interior coincidence the fibre

is *exactly* the Stanley–Reisner scheme of m points times the weight line. This upgrades theorem 7 from the tangent cone to the fibre itself, and since the scheme is reduced there are no nilpotent directions that the second-order test κ could have missed. The simplicial complexes of the gluability filtration in the companion report [31] and the singular fibres of level two are therefore the same kind of object.

Three variables: closed form, flatness and the hyperdeterminant. For $m = 3$ the map $(w_1, \mu, d) \mapsto (\mu, m_{12}, m_{13}, m_{23}, m_{123})$ goes between spaces of dimension 7 and can be inverted. Put $\Delta_m = m_{12}m_{13}m_{23}$ and $\mathrm{Det} = m_{123}^2 + 4\Delta_m$.

Proposition 12 (SC3, SC4)**.** *(i) With* $\{i, j, k\} = \{1, 2, 3\}$,

$$w_1 w_2 = \frac{\Delta_m}{\mathrm{Det}}, \qquad w_2 - w_1 = \pm\sqrt{1 - \frac{4\Delta_m}{\mathrm{Det}}}, \qquad d_i = \frac{m_{123}}{(w_2 - w_1)\, m_{jk}};$$

the sign of the square root is the labelling.

(ii) The Jacobian determinant is $-(w_1 w_2)^3 (d_1 d_2 d_3)^2$.

(iii) Det *is Cayley's hyperdeterminant [17] of the* $2 \times 2 \times 2$ *table of the law.*

Theorem 13 (flattening stratification, $m = 3$; SC5)**.** *Over the space of laws:*

laws	*fibre*	*flat*
all $m_{ij} \neq 0$, $\mathrm{Det} \neq 0$	*two reduced points (labels swap)*	*yes (étale, degree 2)*
one variable independent of the other two	*a surface*	*no*
product laws	*Stanley–Reisner scheme (dim. 3)*	*no*
exactly one $m_{ij} = 0$*; or* $\mathrm{Det} = 0 \neq \Delta_m$	*empty (closure only)*	—

On the parameter side the map is flat exactly off $\{w_1 w_2 d_1 d_2 d_3 = 0\}$*: where an actor is absent or the actors agree on a variable. On the generic stratum the real points are: two genuine mixtures* ($w \in (0, 1)$) *if* $\Delta_m > 0$*; two signed mixtures (real, but* $w \notin [0, 1]$*) if* $\mathrm{Det} > 0 > \Delta_m$*; a complex-conjugate pair if* $\mathrm{Det} < 0$.

Proof. The fibres are read off proposition 12 and checked by Gröbner bases. Flatness: source and target are regular of the same dimension, so by "miracle flatness" [23, Thm. 23.1] the map is flat at a point iff the fibre through it has dimension 0 there; the Jacobian vanishes exactly on $w_1 w_2 d_1 d_2 d_3 = 0$, and those points lie in the positive-dimensional fibres of the second and third rows. Real points: $p = w_1 w_2 = \Delta_m/\mathrm{Det}$ is real and $1 - 4p = m_{123}^2/\mathrm{Det}$. If $m_{123} \neq 0$, w is real iff $\mathrm{Det} > 0$, and then d is real by (i). If $m_{123} = 0$, then $w = (\frac{1}{2}, \frac{1}{2})$ and $d_i^2 = m_{ij}m_{ik}/(p\, m_{jk})$ has the sign of Δ_m, which here is the sign of $\mathrm{Det} = 4\Delta_m$. In both cases $w \in (0, 1)$ iff $p \in (0, \frac{1}{4}]$ iff $\Delta_m > 0$. The sign of Det separates real tensor rank 2 from 3, as in de Silva and Lim [16]. □

This corrects the picture of section 5. The fold where real explanations disappear is *not* where flatness fails: over $\mathbb{C}$ the two explanations persist and become conjugate. What the stratification gives is the following.

Corollary 14 (where explanations can die)**.** *Along a continuous path of laws, a two-type probability explanation (parameters in the box* $w \in [0, 1]$*,* $a \in [0, 1]^{2m}$*,* $m = 3$*) can cease to exist only (a) by reaching a wall of the box, or (b) at a law where its limit lies in the non-flat locus* $\{w_1 w_2 d_1 d_2 d_3 = 0\}$.

Proof. On $\{\Delta_m > 0\}$ the inverse is continuous (proposition 12, $\mathrm{Det} \geq 4\Delta_m > 0$), so an explanation persists while the path stays there and inside the box. If the path leaves $\{\Delta_m > 0\}$ at a law P^* while the explanations stay in the compact box, a limit point x^* has $F(x^*) = P^*$ with $\Delta_m(P^*) = (w_1 w_2)^3 (d_1 d_2 d_3)^2 = 0$. □

The box wall is the semialgebraic part, invisible to the scheme; the inequalities that cut the probability image out of the secant variety are those of Allman, Rhodes, Sturmfels and Zwiernik [12] (see also [26]).

Corollary 15 (monodromy from the stratification)**.** *Over $\{\Delta_m > 0\}$ the sign of d_1 is a continuous, nowhere-zero label of the two sheets, so the real double cover of genuine mixtures is trivial: roles cannot be exchanged without leaving through the non-flat locus (theorem 9). Over $\mathbb{C}$ the cover is connected and its monodromy is $\mathbb{Z}/2$. For $m = 4$, a variable independent of the rest ($d_4 = 0$) is a flat point (full rank, two points; SC6), the non-flat walls move to codimension ≥ 2 and no longer separate the real region, which is why the 4-variable loop of section 6 can exchange roles.*

We expect that for every $m \geq 3$ the map is flat exactly where $w_1 w_2 \neq 0$ and at least three d_i are non-zero (consistent with [4]); we have checked this only for $m = 3$ and at a point for $m = 4$.

The image. The Zariski closure of the image of the k-type model is the k-th secant variety of the Segre embedding of $(\mathbb{P}^1)^m$ [21]. For $k = 2$, $m = 3$ it fills $\mathbb{P}^7$ (generic $2 \times 2 \times 2$ tensors have rank 2), which is why the boundary of the two-type region is real and semialgebraic (the walls $\Delta_m = 0$, $\mathrm{Det} = 0$ and the box), not Zariski-closed. For $m \geq 4$ the secant variety has dimension $2m + 1 < 2^m - 1$ and is a proper subvariety, toric in secant cumulants [24]. Figure 2 shows the three real regions on a slice, the complex Δ, and a path crossing the walls.

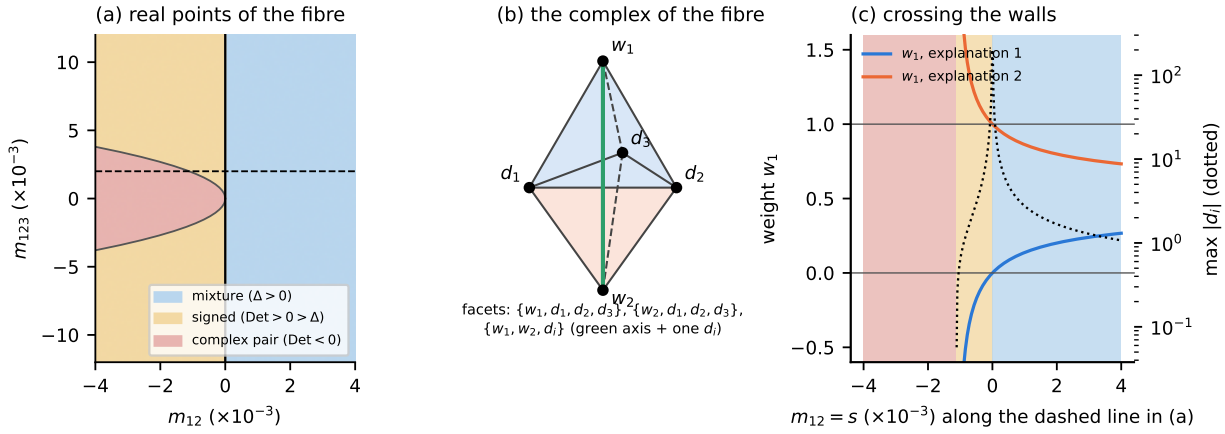


Figure 2: (a) A slice of the law space ($m_{13} = m_{23} = 0.03$, mean fixed) coloured by the real points of the two-type fibre: genuine mixtures ($\Delta_m > 0$), signed mixtures ($\mathrm{Det} > 0 > \Delta_m$), complex-conjugate pairs ($\mathrm{Det} < 0$, inside the parabola $m_{123}^2 = -4\Delta_m$). (b) The simplicial complex Δ of theorem 11 for $m = 3$: two tetrahedra on the triangle $d_1 d_2 d_3$, and the axis $w_1 w_2$ with the triangles $\{w_1, w_2, d_i\}$; its Stanley–Reisner scheme is the fibre over an independent law. (c) Along the dashed line of (a): the weights of the two explanations (they leave $[0, 1]$ as the path crosses $m_{12} = 0$, where one weight tends to 0 and $\max |d_i|$ blows up, and they are complex beyond $\mathrm{Det} = 0$). A probability explanation also needs $|d_i| \leq 1$, so in practice it hits the box wall before $m_{12} = 0$.

8 The ladder of base semirings

The two levels fail in the same way: their equations have points over a larger number system but not over the non-negative one. We make this a single picture. Write the evidence at either level as polynomial equations with rational coefficients, and present a probability a together with its complement $\bar{a} = 1 - a$, so that the parameter box is cut out by $a, \bar{a} \geq 0$. Then ask over which semiring the equations have points:

$$\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\geq 0}.$$

An $\mathbb{R}_{\geq 0}$-point is a probability explanation; an $\mathbb{R}$-point that is not one is a signed explanation (a quasi-probability, or actors with negative weight); a $\mathbb{C}$-point that is not real is an oscillatory explanation; and no point at all is the bottom of the ladder. Each rung has its own measure of how badly the next rung up fails (table 3).

rung	level one (a law on outcomes)	level two (two latent actors, three variables)
none	the tables disagree on an overlap (signalling, a direct influence); measured by the *linear gap*, the least ℓ_1 repair	not in the image: one pairwise moment zero alone, or $\mathrm{Det} = 0 \neq \Delta_m$
$\mathbb{C}$	same as $\mathbb{R}$ (linear equations)	$\mathrm{Det} < 0$: a complex-conjugate pair; measured by the imaginary part
$\mathbb{R}$	contextual: a signed global section exists, no positive one; measured by the *negativity* ν	signed mixture, $\mathrm{Det} > 0 > \Delta_m$; measured by the negative weight $\nu_2 = \frac{1}{2}\left(\lvert m_{123}\rvert/\sqrt{\mathrm{Det}} - 1\right)$
$\mathbb{R}_{\geq 0}$	the tables glue ($\lambda^* = 1$)	genuine mixture, $\Delta_m > 0$; then the box $a \in [0,1]$ is a further semialgebraic wall

Table 3: The ladder at both levels. Level one has no complex rung, because its equations are linear; level two has one, because they are not.

Level one. Let $A\mu = t$ be the marginal equations of a family of windows, with μ a signed measure of total mass 1. The *linear gap* is $\min \|A\mu - t\|_1$, and when it is zero the *negativity* is $\nu = \min\{\|\mu^-\| : A\mu = t\}$, the least total negative mass of a signed global section; both are linear programs. The negativity is the quantity behind Abramsky and Brandenburger's operational reading of negative probabilities [2].

Theorem 16 (the ladder at level one; LD1–LD3)**.** *(i) $\mathbb{C}$-points exist iff $\mathbb{R}$-points exist iff the linear gap is 0; for the measurement scenarios we test (cycles, CHSH, three outcomes) this holds whenever the tables agree on their overlaps [1].*

(ii) $\mathbb{R}_{\geq 0}$-points exist iff $\nu = 0$ iff $\lambda^* = 1$.

(iii) On the set of families with linear gap 0 (a polytope), ν and $1 - \lambda^*$ are comparable: there are constants $0 < c_- \leq c_+$ depending only on the scenario with $c_-(1 - \lambda^*) \leq \nu \leq c_+(1 - \lambda^*)$. One can take $c_+ = \nu(r)$ maximised over the no-signalling polytope.

Proof. (i) The equations are linear with rational coefficients, so solvability over $\mathbb{C}$, $\mathbb{R}$ and $\mathbb{Q}$ coincide; the gap is the ℓ_1 distance to solvability. (ii) A signed section with $\nu = 0$ is a global law, and conversely. (iii) Upper bound: write $t_w = \lambda^* q|_w + (1 - \lambda^*) r_w$ as in (1). The residues r_w agree on overlaps because t and $q|_w$ do, so r has a signed section μ_r, and $\lambda^* q + (1 - \lambda^*)\mu_r$ is a signed section of t with negativity $(1 - \lambda^*)\nu(\mu_r)$. Lower bound: ν and $1 - \lambda^*$ are both convex and piecewise linear in t (values of linear programs with t in the right-hand side), and they vanish on the same polytope,

the noncontextual polytope. A polyhedral convex function grows at least linearly away from its zero set (weak sharp minima [6], a form of Hoffman's bound [28]), and the other is Lipschitz and vanishes there; so each is bounded by a multiple of the distance to that polytope, and hence by a multiple of the other. □

The constants depend on the scenario, and we measured them (fig. 3(a), 30 random mixtures of a global law with a contextual family in each of five scenarios). The ratio $\nu/(1-\lambda^*)$ is exactly $\frac{1}{2}$ for the 3-cycle, the 4-cycle and CHSH [14] with the Popescu–Rohrlich box [10], exactly $\frac{1}{4}$ for the anti-correlated 5-cycle, and between $\frac{1}{2}$ and $\frac{2}{3}$ for a three-outcome triangle, never exceeding the contextual part's own negativity. So the negativity is not a fixed multiple of the unsavable fraction; the two measure the same thing, but in different units.

The drug example sits on the bottom rung. In example 3 the B margin differs between the two populations ($P(H_B)$ is 0.26 in one and 0.40 in the other), so the pair tables disagree on an overlap: there is no signed global law at all, and the linear gap is 0.28. Yet $\lambda^* = 0.86$: the graded explanations of section 2 are about sub-measures, and they still save 86% of the structure. In the language of contextuality-by-default [27] this is a direct influence, not contextuality; the ladder separates the two, and λ^* measures both together. Population 1 alone is on the top rung.

Level two. For two actors and three variables, theorem 13 already places every generic law on a rung. The measures follow from proposition 12: since $1 - 4w_1w_2 = m_{123}^2/\mathrm{Det}$, the weights are $\frac{1}{2}(1 \pm |m_{123}|/\sqrt{\mathrm{Det}})$, so on the $\mathbb{R}$ rung the negative weight is $\nu_2 = \frac{1}{2}(|m_{123}|/\sqrt{\mathrm{Det}} - 1)$, which tends to 0 as $\Delta_m \to 0^-$ and to ∞ as $\mathrm{Det} \to 0^+$. On the $\mathbb{C}$ rung the imaginary parts measure the failure; along the Puiseux path of section 5 the explanation for $s < 0$ is $c \pm i\sqrt{|s|}\,e$, with imaginary part $2\sqrt{|s|}\max_i |e_i|$ in d.

Proposition 17 (how a law moves between rungs; LD4–LD6)**.** *For $m = 3$ a two-type explanation moves between rungs only (a) across $\mathrm{Det} = 0$, where the weights diverge, (b) across $\Delta_m = 0$, either through a non-flat stratum or through a point of the closure that is not in the image, where one weight passes through 0 while $\max|d_i|$ diverges, or (c) across a wall of the box. In (a) and in the second case of (b) the fibre has a point at infinity: the map is not proper there.*

Proof. The rung is determined by the signs of Det and Δ_m and by the box (theorem 13). By proposition 12, $w_1w_2 = \Delta_m/\mathrm{Det}$ diverges as $\mathrm{Det} \to 0$ with $\Delta_m \neq 0$; and $\Delta_m \to 0$ with $\mathrm{Det} \to m_{123}^2 \neq 0$ sends one weight to 0 and, if only one pairwise moment vanishes, $d_k = m_{123}/((w_2 - w_1)m_{ij})$ to infinity. □

The checks confirm the closed-form rung against how each law was made (200 genuine mixtures, 200 valid laws made from signed weights, and complex cases confirmed by an exact Gröbner solve; about 8% of laws drawn at random from a Dirichlet distribution lie on the complex rung), and follow a path across all the walls (fig. 3(b)).

What the ladder says. Incoherence comes in kinds, and the kinds are the rungs. Disagreement on an overlap is the bottom rung: no number system rescues it, and the only remedy is to move the evidence (the consistency radius, the linear gap). Contextuality is the $\mathbb{R}$ rung at level one: the evidence is consistent as linear algebra but needs negative probability. At level two the same rung means actors with negative weight, and a new rung appears below it: actors that exist only as complex conjugates. The gradation of section 2 cuts across the ladder, since λ^* (or λ_2^*) measures how much of the evidence can be saved on the top rung whatever rung the whole sits on. The scheme fixes the equations; the semiring decides whether their points are probabilities. A single

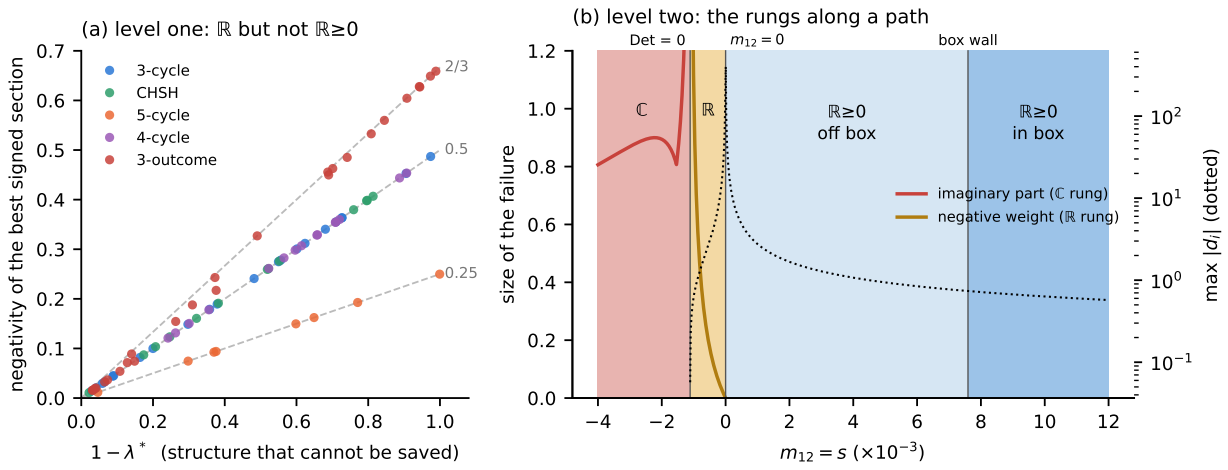


Figure 3: (a) Level one: the negativity of the best signed global section against $1 - \lambda^*$ for no-signalling families in five scenarios; each lies on or between lines whose slope is its contextual part's own negativity. (b) Level two, along the path $m_{12} = s$ ($m_{13} = m_{23} = 0.03$, $m_{123} = 0.002$, means $\frac{1}{2}$): the rung of the two-type explanation, the imaginary part on the $\mathbb{C}$ rung, the negative weight on the $\mathbb{R}$ rung, and $\max|d_i|$ (dotted), which diverges at $m_{12} = 0$. The explanation enters the box at $s \approx 0.0076$.

semiring scheme carrying both levels, with λ^* read from it, is the step this section motivates but does not take.

9 Obstructions to commutativity as structure

Everything so far is commutative: probabilities are numbers, and the order in which evidence is taken does not matter. Here we let the coefficients be matrices. A state is a density matrix ρ (Hermitian, positive semidefinite, trace 1), an observable is a projective measurement $\{A_a\}$ (orthogonal projections summing to the identity), and spatial arrangement enters through rotations: the spin-j component $n \cdot J$ along a direction n. The aim is not to discard what fails to commute but to give it a place. It finds three: the cover of contexts, the complex rung of the ladder, and the holonomy of loops. All statements are checked in `engine/check_noncomm.py` (NC1–NC8).

Commutators decide the cover. Observables that commute can be measured together. The contexts are therefore the maximal sets of pairwise commuting observables (cliques of the commutation graph), and a state ρ assigns each context its joint table $\mathrm{tr}(\rho \prod_i A_{a_i}^{(i)})$. These tables agree on overlaps automatically. If all observables commute there is one context and the tables glue (the joint spectral measure; NC1). Gluing can fail only through non-commutativity. For the CHSH observables the commutation graph has exactly the four Bell contexts, and at Tsirelson's angles [48] the engine's noncontextual fraction is $\lambda^* = 2 - \sqrt{2}$, the known value. As the angle between A_0 and A_1 opens from 0, $\|[A_0, A_1]\|$ grows from 0 and λ^* falls from 1 (NC3, fig. 4(b)). Classically the cover was given; here it is derived from the algebra, and the gluing questions of sections 2 and 8 sit on top of it.

Order is bounded by the commutator. Measuring P then Q (Lüders' rule [40]) gives $\mathrm{tr}(QP\rho PQ)$, so the order effect is $O = \mathrm{tr}\big(\rho\,(PQP - QPQ)\big)$.

Proposition 18 (NC2). *$|O| \le \|[P, Q]\|$ for every state, and the QQ equality $p(P, \bar{Q}) + p(\bar{P}, Q) = p(Q, \bar{P}) + p(\bar{Q}, P)$ holds for every pair of binary projective measurements.*

Proof. By the two-subspace theorem [37] the space splits into blocks where P and Q commute (contributing 0) and 2×2 blocks with an angle θ, where $P = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and Q projects onto $(\cos\theta, \sin\theta)$. There $PQP - QPQ = \cos^2\theta\,(P - Q)$, of norm $\cos^2\theta\,|\sin\theta| \le |\cos\theta \sin\theta| = \|[P, Q]\|$ on the block. The QQ equality follows by expanding with $\bar{P} = 1 - P$, $\bar{Q} = 1 - Q$: $P\bar{Q}P + \bar{P}Q\bar{P} = P + Q - PQ - QP$, which is symmetric in P and Q, so both sides equal $\mathrm{tr}\big(\rho(P + Q - PQ - QP)\big)$. □

The QQ equality is the test Wang, Solloway, Shiffrin and Busemeyer applied to order effects in survey questions [49]; it holds for any projective model and constrains the order effects that commutation leaves.

The canonical section lives on the complex rung. For two measurements A and B the Kirkwood–Dirac table [38, 36, 33]

$$q(a, b) = \mathrm{tr}(B_b A_a \rho)$$

has exactly the right marginals, $\sum_b q(a, b) = \mathrm{tr}(A_a \rho)$ and $\sum_a q(a, b) = \mathrm{tr}(B_b \rho)$: it is a global section of the two tables. But its entries are complex, and

$$\mathrm{Im}\, q(a, b) = \tfrac{1}{2i}\,\mathrm{tr}\big([B_b, A_a]\rho\big),$$

so the imaginary part is an expectation of commutators (NC4). If A and B commute, $B_b A_a$ is a projection and q is a law: the top rung. Real projections with a real state give a real q that can be negative: the $\mathbb{R}$ rung. Otherwise q is non-real: the $\mathbb{C}$ rung. At level one, with numbers as coefficients, the complex rung was empty (table 3); with matrices it is inhabited, and its inhabitants

are exactly the commutation obstructions. This is a finer statement than λ^*. Two contexts that do not overlap always glue classically (take the product), so $\lambda^* = 1$ there; what fails is that the section the state itself determines is not a law. Pusey showed that anomalous weak values, which are ratios of Kirkwood–Dirac entries, are proofs of contextuality [46].

Loops leave a phase, and the phase can be non-abelian. Around a loop of rank-one projections the product $\mathrm{tr}(P_1P_2P_3)$ (the Bargmann invariant [34]) is gauge-invariant. For spin $\frac{1}{2}$ its argument is half the signed solid angle of the spherical triangle, the Pancharatnam–Berry phase [44, 35]; for real projections it is real (NC5, fig. 4(d)). Complex coefficients are therefore what makes loops carry information: the phase is invisible in each probability and appears only in the order of the product. For spin j, densely sampled loops return the state with phase $j\Omega$ (NC8), so two paths to the same target with the same success probability differ by a geometric phase. With rank-two projections the loop acts on the subspace by a 2×2 unitary, and two different loops give unitaries that do not commute (NC6): the holonomy is non-abelian [50]. This is where the $\mathbb{Z}/2$ exchange of actors in section 6 goes once actors carry matrices: the discrete swap becomes one element of a continuous, non-abelian holonomy.

Probability paths. Projections that do not commute can steer. Start in $|j, j\rangle$ along n_0 and measure $n \cdot J$ along N equally spaced directions on the great circle to n_1 (angle Θ). Each step keeps the tracked outcome with probability $|d^j_{jj}(\Theta/N)|^2 = \cos^{4j}(\Theta/2N)$, so

$$P_N = \cos^{4jN}\Big(\frac{\Theta}{2N}\Big) \;\longrightarrow\; 1 \qquad (N \to \infty),$$

against $\cos^{4j}(\Theta/2)$ for the direct jump (NC7, exact for $j = \frac{1}{2}, 1, \frac{3}{2}$; fig. 4(c)). This is Zeno dragging [42, 32]. Projections that commute with each other and with the state can only condition; they never move it (NC7). Read as an analogy for molecular arrangements: a configuration that is improbable in one step can be reached with probability close to one along a path of intermediate arrangements, the route matters through the geometric phase even when the success probability does not, and the whole effect lives in the non-commutativity. Whether a given chemical system behaves this way is a physical question these toy spin models do not answer.

What survives of the scheme picture. For a fixed matrix dimension d, take the real and imaginary parts of the entries as coordinates: marginals (partial traces) are linear, Born probabilities quadratic, and $\rho \succeq 0$ a set of polynomial inequalities. Explanations are again real points of a commutative scheme, and the strata become rank conditions. Three things change. The positive cone is the positive semidefinite cone, which is not a semiring, so the ladder of section 8 becomes a ladder of cones, classical $\subset$ quantum $\subset$ signed (between local and no-signalling sits the quantum set, approximated by the NPA hierarchy [43]). Actors carrying matrices have a unitary gauge freedom, so fibres become orbits and the right object is the quotient, whose coordinate ring is generated by trace polynomials [45]. And if d is not fixed there is no single scheme: the observables generate a non-commutative algebra whose representation schemes Rep_d form a tower [39]. Whether some d explains the data is the question behind Tsirelson's problem, and it is undecidable in general [47, 41]. Our classical results are the diagonal sector of this picture. Whether the fibre over a merge point stays reduced once off-diagonal (coherence) directions are allowed is the first open scheme-theoretic question here.

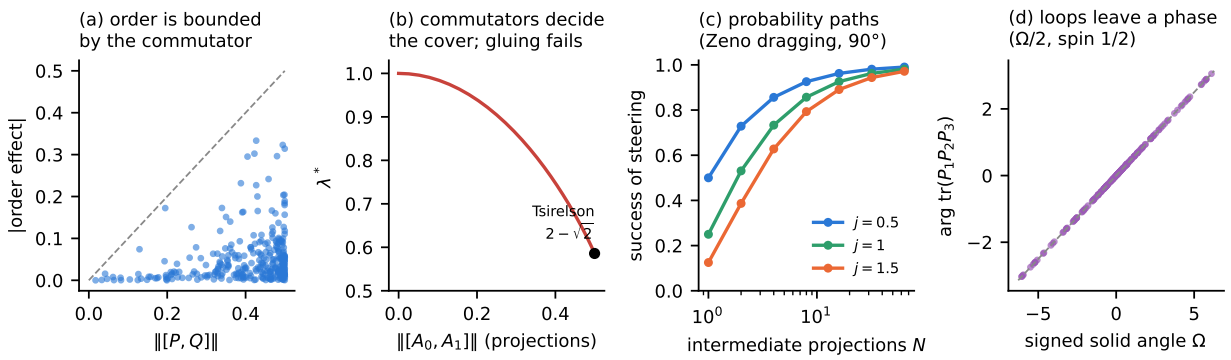


Figure 4: (a) Order effects against $\|[P,Q]\|$ for 300 random pairs of projections: all below the diagonal (proposition 18). (b) The CHSH family: as the angle between A_0 and A_1 opens, the commutator grows and λ^* falls from 1 to $2-\sqrt{2}$. (c) Zeno dragging through 90°: success against the number of intermediate projections, for spin $j = \frac{1}{2}, 1, \frac{3}{2}$. (d) The loop phase $\arg\operatorname{tr}(P_1P_2P_3)$ against the signed solid angle for 200 random spin-$\frac{1}{2}$ triangles: the line $\Omega/2$.

10 Given measurements, partly masked

Section 9 ended at a fork: either the measurements are given, or they too are unknown, in which case deciding whether some dimension explains the data is undecidable in general. Here we take the measurements as given and then hide a fraction of them, to see how the difficulty grows and how much geometry and positivity still recover. At a fixed dimension nothing becomes undecidable: the questions are first-order statements over the reals and so decidable by quantifier elimination [51]. What grows is the cost, the non-convexity, and the size of the set of explanations. All results are in `engine/check_masked.py` (MK1–MK6) and `engine/check_masked_gauge.py` (MK7).

Setting. Two qubits; Alice measures spin along $K = 5$ known directions a_i, Bob along $K = 5$ known directions b_j; every pair is a context whose 2×2 table is observed with $N = 2000$ shots, from a random entangled state mixed with 15% noise. A masked direction still has its tables observed, but the solver does not know it. With Bloch vectors s, t and correlation matrix T,

$$P(\alpha,\beta \mid a,b) = \tfrac{1}{4}\big(1 + \alpha\, s\cdot a + \beta\, t\cdot b + \alpha\beta\, a^{\top}Tb\big),$$

linear in the state for fixed directions and linear in each direction for a fixed state. With nothing masked the state is a linear least-squares fit, unique and positive within noise (MK1).

What the linear fit can see.

Proposition 19 (MK5, MK7)**.** *Let Alice's known directions span $V \subseteq \mathbb{R}^3$ and let h_A of her directions be masked; let Bob's directions span $\mathbb{R}^3$. Then the tables do not change if each hidden direction n is replaced by Mn and the state by $s \mapsto M^{-\top}s$, $T \mapsto M^{-\top}T$, for any $M \in GL_3$ that fixes V pointwise. The linear fit therefore leaves the hidden directions ambiguous along this group, cut by unit length: a family of dimension $\max\big(0,\, 3(3-\dim V) - h_A\big)$. In particular the hidden directions are located by the linear fit iff the known directions on their side span $\mathbb{R}^3$; and with three directions per side and $\dim V = 2$, the fit is the same for every candidate direction.*

Proof. Alice's data enter only through $s\cdot a$ and $a^{\top}Tb$. For a known $a \in V$, $(M^{-\top}s)\cdot a = s\cdot M^{-1}a = s\cdot a$ and likewise for T; for a hidden n, $(M^{-\top}s)\cdot Mn = s\cdot n$. The group of M fixing V pointwise

has dimension $3(3 - \dim V)$, and unit length imposes one condition per hidden direction. The dimension count is checked as the kernel of the Jacobian at the truth in seven cases ($\dim V = 3, 2, 1$; $h_A = 1, 2, 3$), all equal to the prediction (MK7). With three directions per side the nine tables carry exactly the fifteen state parameters, so for planar known directions the residual is zero at 92% of lattice points (the rest lie near the plane, where the three directions become dependent; MK5). □

When the known directions span $\mathbb{R}^3$, the linear fit has one basin: refining from the best lattice point recovers a hidden direction exactly on noiseless data (fig. 5(c)).

What the cone adds: culling impossible points. When the linear fit is blind, positivity is not. Every candidate direction implies a fitted state; if that state has a negative eigenvalue, it assigns negative probability to some measurement, and the candidate is not merely improbable but impossible. On a lattice this is one eigenvalue per point. In the planar case (two known directions in a plane and one hidden out of it, on each side) only 0.01% of the 40,000 lattice points imply a positive state; none of them is the truth, because the feasible set is a small region that a 14° lattice misses. Using the smallest eigenvalue as a score, keeping the best lattice points, and climbing from them finds the feasible set, which contains the truth, is closed under the mirror (global complex conjugation, which preserves positivity), and bounds invariants: over 1434 feasible samples the CHSH maximum [52] lies in $[2.254, 2.452]$ (truth 2.354), and a held-out correlation in $[0.159, 0.256]$ (truth 0.211) (MK6, fig. 5(d)). A mirror applied to one side only is a partial transpose, which does not preserve positivity for entangled states [53], so the cone removes one-sided ambiguities while keeping the two-sided one.

Random masking. Table 4 summarises six random trials at each level (fresh state and directions each time). Hiding one or two of ten directions always leaves at least three known on each side, so the linear fit locates them: an exhaustive lattice of 300 directions per hidden direction is culled to under 2% by the fit alone, and in every trial whose known directions are well-conditioned (9 of 12) exactly one explanation survives the cone (MK2). The exception is instructive. In one trial at 20%, with nearly coplanar known directions, the linear fit left three explanations and none of them implies a positive state, although the truth lies inside the χ^2 set: when the geometry is nearly degenerate the positive explanation sits away from the linear optima, and positivity has to drive the search, as in the planar case above, rather than be applied afterwards. At 30% and 50% the exhaustive lattice would need 300^h fits, so coordinate-wise lattice sweeps take over; they reach the best explanation found by any method (MK3). The errors are large exactly in the trials where a side keeps fewer than three independent known directions, as proposition 19 predicts.

masked	located linearly	median error	truth in 99% set	fits used	local-search success
0%	6/6	0°	6/6	1	—
10% (1 of 10)	6/6	3.0°	5/6	300	0.94
20% (2)	6/6	3.5°	6/6	90,000	0.81
30% (3)	4/6	4.4° (others 67°)	6/6	16,200	0.74
50% (5)	0/6	(all 49°)	6/6	27,000	0.69

Table 4: Masked measurements, six trials per level. "Located linearly": each side keeps known directions spanning $\mathbb{R}^3$. The error is the largest angle between a recovered and a true hidden direction. Fits: exhaustive lattice for one and two hidden directions, lattice sweeps beyond. Local search: the fraction of random starts of plain least squares that reach the best explanation.

Three further points. The coverage of the 99% χ^2 set was 11 of 12 at 10–20%, so the Gaussian approximation behind it is somewhat optimistic. The operator-free rungs of section 8 are computed from the tables alone, so masking does not touch them: the noncontextual fraction of the tables was between 0.947 and 0.967 in the sampled trials, whatever was hidden (MK4). And what can be extracted is exactly what is invariant under the residual symmetry: the CHSH maximum survives the mirror and is pinned by the cone, while a correlation with a new direction is not invariant and has a wider interval. In the language of section 9, what masked measurements leave recoverable are the functions on the quotient by the residual group, cut down by positivity.

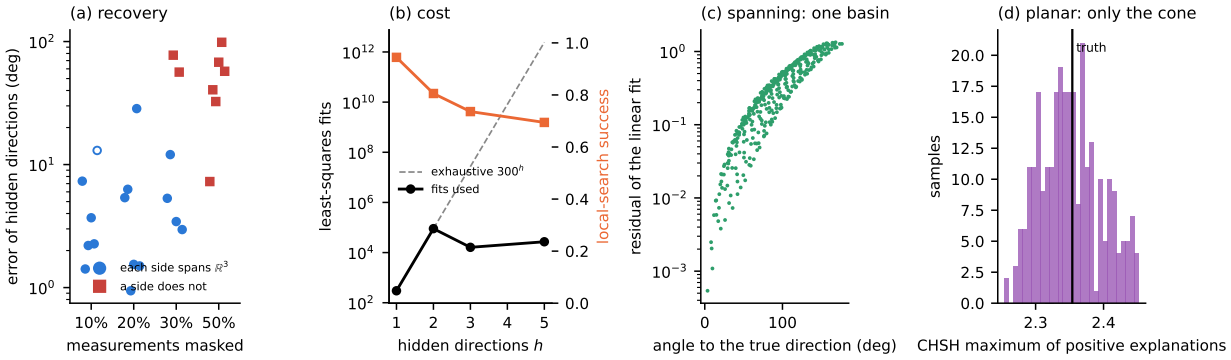


Figure 5: (a) Error of the recovered hidden directions against the masking level, split by whether each side keeps known directions spanning $\mathbb{R}^3$ (circles) or not (squares); an open marker is a trial whose truth fell outside the 99% set. (b) Cost: least-squares fits used (exhaustive lattice up to two hidden directions, sweeps beyond) against the exhaustive count 300^h, and the success rate of plain local search. (c) Spanning known directions: the residual of the linear fit over the lattice against the angle to the true direction: one basin. (d) Planar known directions: the linear fit is blind, the cone is not; the CHSH maximum over the feasible (positive) explanations against the truth.

11 Quantum graded gluing: positivity domains of hereditary quadratics

Islam, Volčič and Woerdeman study self-adjoint matrix polynomials $f = L + H^*H - K^*K$ in non-commuting variables x_j and their adjoints, evaluated on tuples of matrices of every size, and their positivity domains $\mathcal{D}(f)$, where f is positive semidefinite [54]. Graded gluing with quantum windows is exactly of this form. All statements here are checked in `engine/check_qglue.py` (QG1–QG7).

The identification. Let the windows be sets of qubits with density matrices ρ_w, and let ψ be an unnormalised global vector. The quantum form of $\alpha\, q|_w \le t_w$ is

$$\mathrm{Tr}_{\mathrm{rest}}(\psi\psi^*) \preceq \rho_w \quad \text{for every window,} \qquad \text{with mass } \alpha = \|\psi\|^2.$$

Ordering ψ as ψ_{ak} (window index a, the rest k) and putting $K = \sum_{a,k} |k\rangle\langle a|\, x_{ak}$, the constraint is $f_w(\psi) \succeq 0$ for $f_w = \rho_w^{\top} - K^*K$, a self-adjoint hereditary quadratic with constant term and no linear term. The family gives $f = \bigoplus_w f_w$, and the quantum explanations are its positivity domain. The paper's hypotheses hold (QG1): the quadratic coefficients generate the full matrix algebra

(coefficient-irreducible); the line system is minimal; the linearisation $\left(\begin{smallmatrix} I & K \\ K^* & \rho_w^\top \end{smallmatrix}\right)$ generates the full algebra, so it is an irreducible pencil and f_w is an atom; $f_w(0) = \rho_w^\top \succ 0$ exactly when ρ_w has full rank; f_w is unsignatured (positive near 0, indefinite far out); every window is active somewhere (positivity-minimal, QG2). The grading is what makes this work. The normalised problem has an equality $\mathrm{tr} = 1$, which the framework does not allow; the graded problem has inequalities only, and $\psi = 0$ is an interior point.

The walls are algebraic. By Prop. 2.21 of [54] the boundary of $\mathcal{D}_n(f)$ lies in the real free locus $\bigcup_w \{\det f_w = 0\}$, the set where some window's constraint saturates in some direction. By their Theorem 3.18, under the hypotheses above, the boundary is Zariski dense in the free locus at every level that is a multiple of some n_0. At level one we checked the local version: at the boundary point of every ray, the active window has $\det f_w = 0$ and the real gradient of $\det f_w$ is non-zero, so the boundary contains regular real points and the real locus is dense in that component (their Prop. 3.14; QG2). Figure 6(a) shows a slice. Because f has no linear term, $\psi \in \mathcal{D}$ implies $s\psi \in \mathcal{D}$ for $0 \le s \le 1$: the domain is star-shaped, and the topology changes of the kind in their Example 4.3 cannot occur in $\mathcal{D}(f)$ itself. They can only occur in its normalised slices (the quantum E_α), which we have not established here.

Theorem 20 (classical limit and dephasing bound; QG3, QG4)**.** *Let* $\lambda_Q = \max\{\mathrm{tr}\,\sigma : \sigma \succeq 0,\ \mathrm{Tr}_{\mathrm{rest}}\sigma \preceq \rho_w\ \forall w\}$ *(a semidefinite program). For every product basis,* $\lambda_Q \le \lambda^*$ *of the tables obtained by measuring every qubit in that basis. If every* ρ_w *is diagonal,* $\lambda_Q = \lambda^*$*.*

Proof. Dephasing $\mathcal{P}$ in a product basis is positive and order-preserving, and it commutes with partial traces. If σ is feasible then $\mathcal{P}(\sigma)$ is a diagonal operator with $\mathrm{Tr}_{\mathrm{rest}}\mathcal{P}(\sigma) = \mathcal{P}_w(\mathrm{Tr}_{\mathrm{rest}}\sigma) \preceq \mathcal{P}_w(\rho_w)$, so its diagonal is a classical sub-measure whose marginals are at most the tables, and its mass is $\mathrm{tr}\,\sigma$. Hence $\mathrm{tr}\,\sigma \le \lambda^*$. If the ρ_w are diagonal, every classical sub-measure is a feasible diagonal σ. □

So the quantum rung sits below every classical view of the same evidence. On random quantum families it is strictly below: 0.004 to 0.023 below the best of twenty random product bases (fig. 6(c)).

Theorem 21 (the Werner triangle; QG4, QG7)**.** *Let three qubits have all three pairwise windows equal to the Werner state* $W_p = p\,|\Psi^-\rangle\langle\Psi^-| + (1-p)I/4$*,* $-\frac{1}{3} \le p \le 1$*. Then*

$$\lambda_Q = \lambda_Z^* = \min\left(1, \tfrac{3}{2}(1-p)\right),$$

where λ_Z^* *is the classical fraction of the tables measured in the Z basis. The evidence glues fully exactly when* $p \le \frac{1}{3}$*. For* $p \ge \frac{1}{3}$ *an optimal explanation is* $\frac{3}{8}(1-p)\,\Pi_{1/2}$*, supported on the spin-*$\frac{1}{2}$ *sector.*

Proof. Reduction by symmetry. The windows are identical and invariant under $U \otimes U$ and under swapping the two qubits, so the feasible set of the semidefinite program is invariant under $U^{\otimes 3}$ and under permutations of the qubits, and so is the objective. Averaging a feasible σ over this group keeps it feasible and keeps its trace; hence the optimum is attained by an invariant σ. By Schur–Weyl duality the invariant operators are combinations of the isotypic projectors of $(\mathbb{C}^2)^{\otimes 3} = \mathrm{spin}\,\frac{3}{2} \oplus 2 \cdot \mathrm{spin}\,\frac{1}{2}$ (the totally antisymmetric part is zero), so $\sigma = a\,\Pi_{3/2} + b\,\Pi_{1/2}$ with $a, b \ge 0$ and $\mathrm{tr}\,\sigma = 4a + 4b$.

Partial traces. $\mathrm{Tr}_3\Pi_{3/2}$ is supported on the triplet, invariant, and of trace 4, so it equals $\frac{4}{3}\Pi_{\mathrm{trip}}$; and $\mathrm{Tr}_3\Pi_{1/2} = \mathrm{Tr}_3 I - \frac{4}{3}\Pi_{\mathrm{trip}} = \frac{2}{3}\Pi_{\mathrm{trip}} + 2\Pi_{\mathrm{sing}}$. The same holds for every pair.

The linear program. W_p has eigenvalue $\frac{1+3p}{4}$ on the singlet and $\frac{1-p}{4}$ on the triplet, so the constraints are $\frac{4a+2b}{3} \le \frac{1-p}{4}$ and $2b \le \frac{1+3p}{4}$. The first gives $4a + 4b \le \frac{3}{4}(1-p) + 2b$, maximised by the largest admissible $b = \min\left(\frac{1+3p}{8}, \frac{3}{8}(1-p)\right)$. For $p \ge \frac{1}{3}$ the second term is smaller: $a = 0$ and $\lambda_Q = \frac{3}{2}(1-p)$. For $p \le \frac{1}{3}$: $b = \frac{1+3p}{8}$, $a = \frac{2-6p}{16} \ge 0$, and $\lambda_Q = 1$.

The classical fraction. In the Z basis each pair is equal with probability $\frac{1-p}{2}$ (each equal cell $\frac{1-p}{4}$). Every assignment of three bits has at least one equal pair, so for any sub-measure μ below the tables, $\lambda = \sum_x \mu(x) \le \sum_{\text{pairs}} \mu(\text{pair equal}) \le 3 \cdot \frac{1-p}{2}$. For $p \ge \frac{1}{3}$ the bound is attained by mass $\frac{1-p}{4}$ on each of the six assignments with exactly one equal pair; for $p \le \frac{1}{3}$ the mixture of the uniform law on those six (weight $\frac{3}{4}(1+p)$) and on $\{000, 111\}$ is a global law with the given tables. □

The proof explains the numbers. Three pairs cannot all be strongly singlet-like (monogamy), and the best quantum explanation lives in the spin-$\frac{1}{2}$ sector, whose pair marginals carry the most singlet weight. The classical bound mirrors the quantum one: the equal cells 00 and 11 lie in the triplet, whose eigenvalue $\frac{1-p}{4}$ is exactly the probability of each equal cell, and the pigeonhole count over equal pairs is the classical shadow of the triplet constraint. The semidefinite program agrees with the formula at every p tested, each value certified by the dual, operators $Y_w \succeq 0$ with $\sum_w Y_w \otimes I \succeq I$ and $\sum_w \operatorname{tr}(Y_w \rho_w) = \lambda_Q$ (duality gap below 10^{-9}, QG5); this dual certificate plays the role of the Farkas certificate of level one.

One pure actor is not a mixture. The paper's level one is a single pure vector ψ. Its best fraction (found by projected gradient ascent, so a lower bound) is far below the mixed fraction: at least 0.51 against 1 at $p = 0.2$, 0.40 against 0.75 at $p = 0.5$, 0.18 against 0.30 at $p = 0.8$ (fig. 6(d), QG6). At higher levels the paper's domains are free sets: closed under direct sums, unitary similarity and compression to common invariant subspaces (checked at level two, QG1). Their physical reading for us (operator-valued amplitudes on an environment of dimension n) and their relation to mixed explanations remain to be worked out.

What else the paper offers. Their Theorem 4.9 says that for unsignatured atomic hereditary quadratics with $f(0) \succ 0$, equal positivity domains (at every level) hold exactly when $f = S\tilde{f}S^*$. For one window this says the quantum explanation set determines the window's constraint up to congruence; for a family, which is a direct sum and not an atom, it does not apply as stated. And hereditary-quadratic domains are Γ-convex, with a separation theorem (their Theorem 5.8, from [55]): a point outside is cut off by a hereditary-quadratic certificate, a size-free counterpart of the dual certificate above.

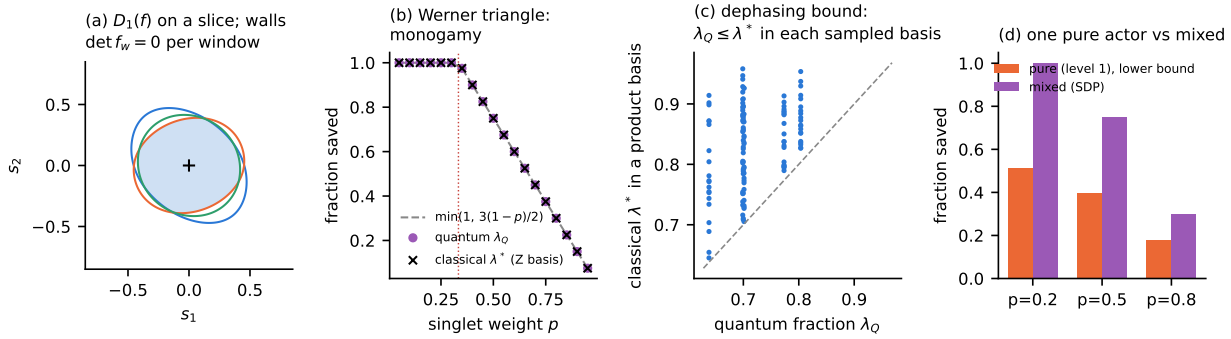


Figure 6: (a) A real two-dimensional slice of the pure domain $\mathcal{D}_1(f)$ for three random pairwise windows on three qubits: star-shaped about 0, bounded by pieces of the walls $\det f_w = 0$ (one colour per window). (b) Pairwise Werner windows: the quantum fraction, the formula $\min(1, \frac{3}{2}(1-p))$ and the Z-basis classical λ^*; full gluing only for $p \le \frac{1}{3}$. (c) The dephasing bound: for six random quantum families, λ^* in twenty random product bases against λ_Q, all above the diagonal. (d) The best pure single-actor fraction (a lower bound) against the mixed fraction for Werner windows.

12 A categorical home

The previous sections left several structures without a common setting: the cover of contexts derived from commutators, the gauge of masked measurements, the quantum rung without a fixed dimension, the information geometry, and the question of which dimension, if any, explains the data. This section places them in one framework. It does not resolve the open problems; it says where each one lives. All computations are in `engine/check_npa.py` (NP1–NP6).

Representation functors and their limits. For an algebra A of observables and a commutative ring B, the representations $\mathrm{Rep}_n(A)(B) = \mathrm{Hom}(A, M_n(B))$ form a functor represented by a commutative affine scheme, with GL_n acting by change of basis; its quotient is described by trace rings [45]. Kontsevich and Rosenberg proposed reading the geometry of A through the family $\mathrm{Rep}_n(A)$ [65], and its derived version is representation homology [57]. Li replaces M_n by an arbitrary finite-dimensional coefficient algebra M, $\mathrm{Rep}_M(A)(B) = \mathrm{Hom}(A, M \otimes B)$, and shows that non-matrix coefficients (strictly upper triangular 3×3 matrices) detect singularities of the quantum plane and the Jordan plane that every matrix coefficient misses [66]. Our fixed-dimension framing of section 9, and the linear gauge of section 10 (the subgroup of GL_3 fixing the known directions), are instances of this functor. What fails at finite dimension is made precise by the following.

Proposition 22 (holonomy against dimension; NP5)**.** *Invertible $n \times n$ matrices with $UV = qVU$ exist iff $q^n = 1$. Hence for $q = e^{2\pi ik/N}$ ($\gcd(k, N) = 1$) they exist exactly when N divides n, and for an irrational phase in no finite dimension; the corresponding algebra (the noncommutative torus) nevertheless has a rich category of infinite-dimensional modules [71, 58]. The Weyl relation $xy - yx = 1$ has no finite-dimensional solution at all.*

Proof. Taking determinants, $\det U \det V = q^n \det V \det U$. Clock and shift matrices tensored with $I_{n/N}$ realise the relation when $N \mid n$. For the Weyl relation take traces: $0 = n$. □

So a loop that carries a phase incommensurable with every finite dimension leaves every Rep_n empty. Numerically, the best unitary residual $\|UV - qVU\|$ is 7×10^{-7} for $q^3 = 1$, $n = 3$, and 1.41, 1.04 for $n = 2, 4$, and stays above 0.75 for an irrational phase (fig. 7(c)). The physical instance is

Tsirelson's problem: Slofstra built games whose perfect quantum strategies exist only in infinite dimensions [73]. This is a different phenomenon from the $\mathbb{Z}/2$ exchange of actors of section 6, which lives at a fixed dimension.

The module category, with positivity. Gabriel and Rosenberg recover a scheme from its category of quasi-coherent sheaves [61, 72], and Artin and Zhang take a category of graded modules as the definition of a noncommutative projective scheme [56]. Taking the category of modules as the space removes the dimension constraint above: infinite-dimensional modules are objects like any other. Two corrections are needed for probability. First, a plain module category has no positivity, and every result of the ladder lives in a cone, so the right category is that of $*$-representations on Hilbert spaces (or of operator systems). Second, a state is not an object but a pointed one: the GNS construction turns a state into a cyclic $*$-representation with a distinguished vector. Observables are recovered as natural endomorphisms of the forgetful functor, and the matrix polynomials of section 11 are morphisms between free modules; their free locus is the set of modules on which such a morphism fails to be invertible.

This is also what makes the framework computable. The NPA hierarchy [43, 68] builds the GNS module from words in the observables, truncated by length, and imposes positivity of its moment matrix; no dimension is fixed. It reproduces Tsirelson's bound $2\sqrt{2}$ at levels 1, $1+AB$ and 2 (NP1), and it gives the dimension-free quantum rung directly on tables: the largest λ with $\lambda q \le P$ cellwise for a quantum behaviour q.

Proposition 23 (the rungs on isotropic boxes; NP2, NP3)**.** *For the box with correlators $\pm v$ and unbiased marginals, $\frac{1}{2} \le v \le 1$: the classical fraction is $\lambda^* = 2(1-v)$, and the quantum fraction is 1 for $v \le 1/\sqrt{2}$ and $(1-v)/(1-1/\sqrt{2})$ beyond.*

Proof. Write the box as $\lambda q + (1-\lambda) r$ with r no-signalling. The CHSH value is linear, equal to $4v$ for the box, at most 2 for local q, at most $2\sqrt{2}$ for quantum q and at most 4 for r; so $4v \le 2\lambda + 4(1-\lambda)$, i.e. $\lambda \le 2(1-v)$, and $4v \le 2\sqrt{2}\lambda + 4(1-\lambda)$, i.e. $\lambda \le (1-v)/(1-1/\sqrt{2})$. Both bounds are attained, by local deterministic boxes and by the Tsirelson box, respectively, mixed with the PR box. □

The engine's linear program and NPA reproduce both formulas at 21 values of v to 4×10^{-10} (fig. 7(a)); on random no-signalling boxes the hierarchy is ordered, $\lambda^* \le \lambda_{\mathrm{NPA}(2)} \le \lambda_{\mathrm{NPA}(1+AB)} \le \lambda_{\mathrm{NPA}(1)} \le 1$ (fig. 7(b)). For two parties with two-outcome measurements, Jordan's lemma reduces quantum correlations to qubits, so here the dimension-free rung adds nothing beyond Rep_2; it matters where proposition 22 bites.

Contexts as a diagram, contextuality as a failed limit. De Silva and Barbosa associate to an operator algebra the diagram of Gelfand spectra of its commutative subalgebras (the spectral presheaf of Isham, Butterfield and Hamilton [63]), and extend any functor on spaces to all algebras by taking the limit over that diagram [59]. The diagram is our commutation cover. Kochen–Specker [64] says its limit of points is empty; Gleason [62] says compatible probability assignments still exist. Contextuality sits between the two: probabilities agree across contexts but come from no measure on global points, and λ^* measures how far.

Proposition 24 (Peres–Mermin; NP6)**.** *For the nine two-qubit Pauli observables of the Peres–Mermin square [69, 67], the commutation cliques are the six lines; no ± 1 assignment respects all six line products; and for every state the six context tables agree on overlaps (a signed global section exists) while $\lambda^* = 0$.*

Proof. Five line products are $+I$ and one is $-I$, so the product of all six constraints over the nine values is $+1 = -1$. Every global assignment therefore violates some line, where it hits a cell of probability zero, so any sub-measure below the tables vanishes. The tables come from one state on jointly measurable sets, so they agree on overlaps. □

Checked for the maximally mixed state and five random states (rung $\mathbb{R}$, $\lambda^* = 0$, linear gap 0). This is state-independent contextuality, the cleanest case where points collapse but probabilities survive.

Information geometry as enrichment. Perrone enriches Markov categories (states as morphisms out of the unit, channels as morphisms) in divergences, so that composition and tensor can only contract them; data processing inequalities follow from the enrichment [70, 60]. For gluing it gives the following.

Proposition 25 (data processing for gluing; NP4)**.** *Applying local channels to the outcomes of each variable never decreases* λ^**.*

Proof. Push the optimal sub-measure μ (with $\mu|_w \leq t_w$) through the channels. Positivity of the channels keeps every marginal below the processed tables, and the mass is unchanged. □

Processing the evidence cannot create contextuality, consistent with the monotones of Abramsky, Barbosa and Mansfield [3]. On ten random boxes the level-$(1+AB)$ quantum fraction never decreased either (numerical, not proved). A quantum Markov category has no copy map, and that failure of copying is non-commutativity itself.

earlier structure	categorical home
cover of contexts (section 9)	the diagram of commutative subalgebras (spectral presheaf)
gluing and its failure (sections 2 and 8)	limits over that diagram: points (Kochen–Specker) versus probabilities (Gleason)
the ladder of rungs	the cone in which the sections live: orthant, positive semidefinite, signed, complex
quantum rung without fixed dimension	positivity on truncated free modules (NPA)
masked directions, gauge (section 10)	the GL_n action on $\mathrm{Rep}_n(A)$ and its quotient
holonomy and phases (section 9)	relations such as $UV = qVU$; finite dimension only when $q^n = 1$
positivity domains (section 11)	morphisms of free modules; free loci as non-invertibility
information geometry	divergence-enriched Markov categories; data processing
level-two latent strata, flattening, hyperdeterminant	remain commutative algebraic geometry
topology of the quantum slices	open

Table 5: What the categorical view gives a home to, and what it does not.

Table 5 summarises the placement. The categorical view unifies the framework; it does not by itself compute anything, which is why the NPA truncation matters, and it leaves the commutative level-two results and the open problems where they were.

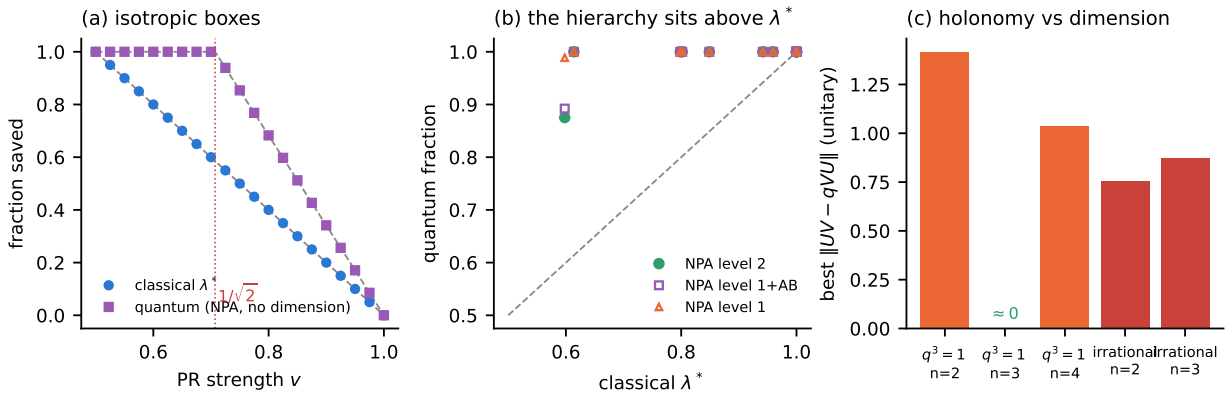


Figure 7: (a) Isotropic PR boxes: the classical fraction $\lambda^* = 2(1-v)$ and the dimension-free quantum fraction (NPA level 1+AB), 1 up to $v = 1/\sqrt{2}$ and $(1-v)/(1-1/\sqrt{2})$ beyond (dashed: proposition 23). (b) Twelve random no-signalling boxes: the quantum fraction at three levels against λ^*. (c) The best unitary residual of $UV = qVU$ by dimension: zero only when $q^n = 1$ (proposition 22).

13 Classifying observations by their origin: a fingerprint, and the Snow Queen data

Nested functors. Each model class $\mathcal{M}$ defines a map from evidence to $[0, 1]$: the largest λ such that a member of $\mathcal{M}$ accounts for a fraction λ of every table at once. The classes are nested, from classical noncontextual (global measures, λ^*), through quantum without fixed dimension (NPA, λ_Q), to no-signalling, and the fractions are ordered, $\lambda^* \le \lambda_Q \le 1$. Each is monotone under local post-processing (proposition 25 for λ^*; checked numerically for λ_Q). Below these sit two diagnostics that decide whether any fraction is meaningful: direct influence s_q (the copies of a variable disagree in their marginals), and the Contextuality-by-Default degree c (incoherence left after the copies are coupled as closely as their marginals allow) [27, 75]. The *origin* of a body of evidence is the lowest rung that explains it: direct influence only ($c = 0$), contextual beyond direct influence ($c > 0$), classical ($\lambda^* = 1$), quantum but not classical ($\lambda_Q = 1 > \lambda^*$), or beyond quantum. The engine computes the fingerprint in one call (`pce/fingerprint.py`; FP1–FP7 check the local, Tsirelson and PR boxes, a signalling local model, a PR box with added direct influence, the level-two labels and the ordering).

These fractions are rudimentary functors in the sense asked for. They need no new algebra; a non-associative coefficient category enters only if the observables themselves compose non-associatively. For Hermitian observables under the Jordan product $A \circ B = \frac{1}{2}(AB + BA)$, the associator is $(A \circ B) \circ C - A \circ (B \circ C) = \frac{1}{4}[B, [A, C]]$, a double commutator (verified numerically). So, for quantum data, the non-associativity of the Jordan product carries no information beyond the commutation cover already used.

The data. Cervantes and Dzhafarov [74] ran a rank-4 cyclic (Bell-type) system on 1,799 respondents. Each respondent saw one context. A context pairs a choice of character (Gerda/Troll, or Snow Queen/Old Finn Woman) with a choice of characteristic (Beautiful/Unattractive, or Kind/Evil). Respondents had to describe a character from the story, where Snow Queen is beautiful and evil. The variables are A_x (character pair x) and B_y (characteristic pair y). The data come in

two sets: D2, correct responses only (their Table 2, $n = 425, 429, 410, 388$), and D3, all responses (their Table 3, $n = 447, 453, 446, 453$).

The analysis was pre-registered and SHA-256-frozen before any response frequency was read (`engine/study8_snowqueen/`). The paper's title already states its conclusion (contextuality), and the registration records this as prior knowledge. One deviation is logged: the frozen script used the total sample sizes for both datasets, and D2 needs the correct-response counts. One cell of Table 3 was illegible in the text extraction. It was fixed from the printed marginals, and the fix was checked by reproducing the authors' published statistics.

Results. The transcribed tables reproduce the authors' Kujala–Dzhafarov statistic exactly: 0.452 on D2 and 0.279 on D3. In both datasets our CbD degree is half of it, $c = 0.226$ and $c = 0.138$. All six predictions held (table 6, fig. 8):

- P1, structural: on D2 every table sits on two cells, with parities $(+, +, +, -)$, so $\lambda^* = \lambda_Q = 0$.
- P2: direct influence is large in both datasets, mostly through Kind/Evil. Its share is $s = 0.47$ on D2 and 0.36 on D3: Kind is chosen far more often when the partner pair is Gerda/Troll.
- P3: on D2, CbD and the Kujala–Dzhafarov criterion agree, and both say contextual.
- P4, the empirical test: on D3 the 95% bootstrap interval of c is $[0.069, 0.194]$, which excludes 0. The data are contextual beyond direct influence, in agreement with the authors' 99.99% interval for their statistic, $[0.008, 0.506]$.
- P5: on D3 the quantum cone accounts for more of every table than the classical one: $\lambda_Q = 0.460$ against $\lambda^* = 0.327$, with gap interval $[0.081, 0.148]$.
- P6: all quantities are invariant under recoding any variable.

	D2 (correct only)	D3 (all responses)
direct influence $\sum_q s_q$	0.774 $[0.718, 0.835]$	0.535 $[0.469, 0.608]$
CbD degree c	0.226 $[0.165, 0.282]$	0.138 $[0.069, 0.194]$
Kujala–Dzhafarov statistic (authors)	0.452	0.279
classical fraction λ^*	0 (structural)	0.327 $[0.280, 0.380]$
quantum fraction λ_Q (NPA 1+AB)	0 (structural)	0.460 $[0.375, 0.515]$
origin	contextual beyond direct influence	contextual beyond direct influence

Table 6: Study 8: the fingerprint of the Snow Queen data, with 95% parametric bootstrap intervals (2,000 replicates).

Reading. The origin is "contextual beyond direct influence" on both datasets, but most of the incoherence is direct influence. The D2 result is largely forced by the design: tables restricted to correct answers are deterministic, and a deterministic cyclic system with odd parity is contextual whenever its direct influence is small enough, a point the authors make themselves. The informative test is D3, and there the contextual remainder is modest, with a lower bound of 0.07.

The gap $\lambda_Q > \lambda^*$ says that a quantum-cone model, as a model class, explains more of the jointly observed evidence than a classical one. It is not evidence for quantum mechanics in cognition: any class whose cone contains the quantum set would do as well, and both fractions are computed on signalling tables, where neither class can explain everything. The next paragraphs separate the

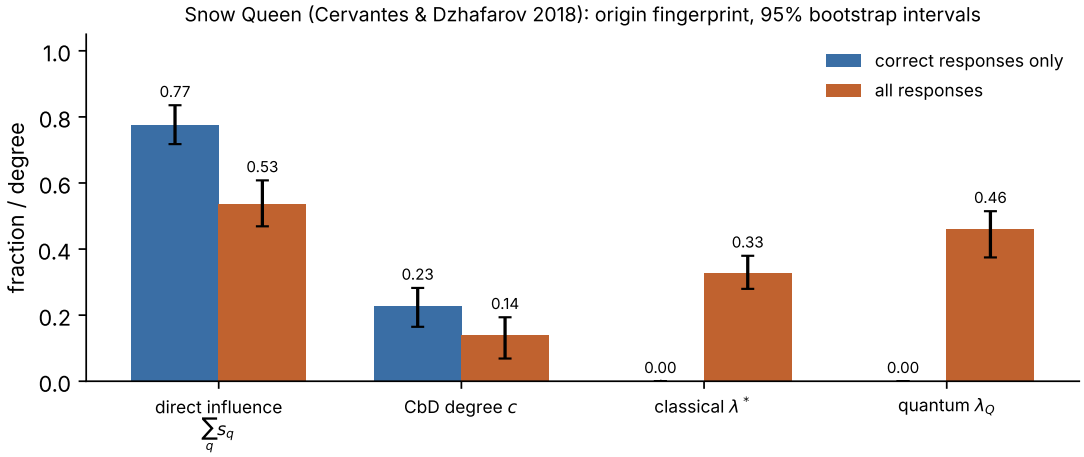


Figure 8: The fingerprint of the two response sets of [74]. Direct influence dominates. A contextual remainder survives sampling in both sets. On all responses, the quantum cone explains 46% of every table jointly and the classical cone 33%.

two. This is the first real-data use of the fingerprint. It is one experiment, published, aggregated, and of known conclusion, so it validates the instrument more than it discovers anything.

Fractions that allow for direct influence. On signalling tables λ^* and λ_Q mix two things: the part of the incoherence that direct influence explains, and the part that only contextuality explains. To separate them we use consistification, in the form used by Camillo and Cervantes for the contextual fraction of inconsistently connected systems [76], within the measures of Kujala and Dzhafarov [77]. Keep one copy of every variable per window, and for every variable seen in two windows add a connection: a further window on its two copies, whose table is their maximal coupling (the copies agree with probability $1 - s_q$, the most their marginals allow). The consistified system has no direct influence, so the ordinary fractions apply to it:

- λ_C^{cbd}, the largest mass of a sub-measure on the copies lying below every window and connection table. This is $1 -$ CNTF, the contextual fraction for inconsistently connected systems [76].
- λ_Q^{cbd}, the same with a quantum model of the copies: ± 1 observables, one per copy, with the cells of every window and connection between 0 and the table, and no fixed dimension (an NPA relaxation over the right-angled Coxeter group of the commutation graph). Copies commute with what they are measured with: their window partner and their other copy. For a Bell-type system we also let every copy of one party commute with every copy of the other.

The second assumption matters. Without it the two copies of a variable can agree on the state without being the same operator, and the quantum set is larger even with no direct influence (CF3). With it, perfectly coupled copies give back the Bell quantum set exactly, by Tsirelson's vector argument: equal vectors $A_x^y \psi = A_x^{y'} \psi$ and commuting parties reproduce a bipartite quantum behaviour.

Proposition 26 (direct influence explains at most its own size)**.** *For windows in which each variable appears at most twice,*

$$\lambda^* \;\geq\; \lambda_C^{\mathrm{cbd}} - \textstyle\sum_q s_q, \qquad \lambda^* \leq \lambda_C^{\mathrm{cbd}} \leq \lambda_Q^{\mathrm{cbd}} \leq 1, \qquad \lambda_C^{\mathrm{cbd}} = 1 \iff c = 0.$$

Proof. Take an optimal sub-measure μ on the copies. Its connection marginals lie below the maximal couplings, so the two copies of q disagree on at most s_q of its mass. Discarding the atoms on which some copies disagree leaves mass at least $\lambda_C^{\mathrm{cbd}} - \sum_q s_q$ on atoms where all copies agree, which is a sub-measure on the original variables below every table. The chain holds because identical copies are admissible in the classical problem, and commuting copies in the quantum one. If $\lambda_C^{\mathrm{cbd}} = 1$ the sub-measure is a coupling of all copies with exact window tables and maximal connections, which is the definition of noncontextuality ($c = 0$), and conversely. □

So $1-\lambda^*$ splits into three non-negative shares: direct influence $\mathrm{DI} = \lambda_C^{\mathrm{cbd}} - \lambda^* \le \sum_q s_q$, quantum-explainable contextuality $\mathrm{QC} = \lambda_Q^{\mathrm{cbd}} - \lambda_C^{\mathrm{cbd}}$, and contextuality beyond quantum $\mathrm{BQ} = 1 - \lambda_Q^{\mathrm{cbd}}$. On cyclic systems the classical share is $1 - \lambda_C^{\mathrm{cbd}} = c$: our CbD degree coincides with CNTF (CF2, on 24 random cyclic systems of rank 3–5). In 240 random cyclic systems the bound on DI was attained in 18%.

The engine checks (`check_cbdfrac.py`, CF1–CF7):

- the relaxation reproduces the n-cycle bound $n\cos(\pi/n)$ for $n = 4, 6, 8$;
- $\lambda_C^{\mathrm{cbd}} = 1 - c$ on cyclic systems;
- with no direct influence, both fractions reduce to λ^* and λ_Q;
- $\lambda_C^{\mathrm{cbd}} = 1$ exactly when $c = 0$, and the ordering holds;
- a quantum model with direct influence has $\lambda_Q^{\mathrm{cbd}} = 1$ while $\lambda_C^{\mathrm{cbd}} < 1$ and $\lambda_Q < 1$. In this model Alice's output is forced to $+1$ through an ancilla, so her copies commute and are maximally coupled;
- a PR box with direct influence gains nothing from quantum models: $\lambda_Q^{\mathrm{cbd}} = \lambda_C^{\mathrm{cbd}}$.

One numerical caveat. Every maximal coupling has a zero cell, so the semidefinite program has no strictly feasible point, and both solvers we tried stop at reduced accuracy. Values reproduce to about 2×10^{-4}. One step of facial reduction did not cure this.

Study 8b: the Snow Queen data with direct influence allowed. The new quantities were registered and frozen as a secondary analysis, with the tables already known. Four logged deviations make the numerics feasible and accurate (`DEVIATIONS.md`): caching the relaxation, switching to Clarabel, a measured solver accuracy of about 2×10^{-4}, and a post-hoc sensitivity analysis. Table 7 and fig. 9 give the decomposition. The registered outcomes:

- R1 and R2 hold: $\lambda_C^{\mathrm{cbd}} = 1 - c$, and $\mathrm{DI} \le \sum_q s_q$ in every replicate. The bound is attained at both point estimates. On these data, direct influence explains exactly as much as its size.
- R3 holds: on the correct-only data the quantum class adds nothing ($\mathrm{QC} = 0$). Its contextual remainder, 0.23, is beyond quantum even with direct influence allowed.
- R4, on all responses, is *undetermined*. The quantum class explains most of the contextual remainder, $\mathrm{QC} = 0.134$ $[0.068, 0.147]$ of $c = 0.138$. The rest is small: $\mathrm{BQ} = 0.005$ at level $1{+}E$ and 0.010 at level 2, with interval $[0, 0.096]$. In 42% of replicates, BQ is below 10^{-4} (43% below 10^{-3}, post hoc), short of the 50% the registered rule needs for "within quantum".
- R5 fails: level 2 is tighter than level $1{+}E$ (0.9903 against 0.9953; level 3 ran out of memory). The registered level is not tight, so the bootstrap values of BQ are lower bounds that may be loose by about 0.005.

- R6 holds trivially, and it is the instructive one. With minimal commutation, $\lambda_Q^{\text{cbd}} = 1$ on both datasets. The conclusion that some contextuality is beyond quantum rests entirely on letting character copies commute with characteristic copies.

	D2 (correct only)	D3 (all responses)
explained classically, λ^*	0	0.327 $[0.280, 0.380]$
direct influence, DI ($= \sum_q s_q$ here)	0.774 $[0.713, 0.831]$	0.535 $[0.467, 0.610]$
contextual, quantum-explainable, QC	0	0.134 $[0.068, 0.147]$
contextual, beyond quantum, BQ	0.226 $[0.169, 0.287]$	0.005 (level 1+E), 0.010 (level 2); $[0, 0.096]$
λ_Q^{cbd} with minimal commutation	1	1

Table 7: Study 8b: $1 - \lambda^*$ split into direct influence, quantum-explainable contextuality and contextuality beyond quantum (bipartite commutation; 95% bootstrap intervals, 2,000 replicates).

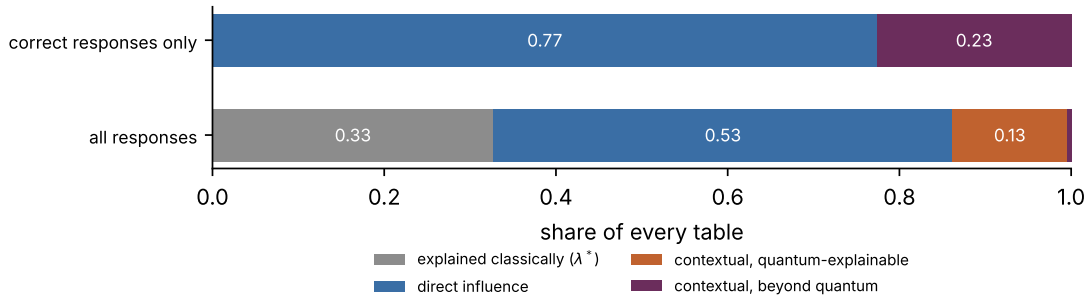


Figure 9: Where the evidence goes. On all responses, a third is explained classically, about half by direct influence, and almost all of the rest by quantum models whose copies may differ. On the correct-only data, quantum models add nothing.

What this adds. Cervantes and Dzhafarov's statistic, and our c, say *that* the all-responses data are contextual. The decomposition says how the incoherence is made up. About 80% of it is direct influence, and no more than the measured size of the direct influence (proposition 26, attained here). Of the contextual remainder, all but 0.005–0.010 is within what a dimension-free quantum model with the same direct influence can produce. Whether the last sliver is real is not decided by these data.

The correct-only data sit beyond quantum, but they are selected on correctness, and a selected sample is not a behaviour any model needs to reproduce. Both statements depend on the commutation assumption, and without it nothing exceeds quantum. As far as we know, this quantum fraction for inconsistently connected systems is new. The classical half is Camillo and Cervantes's CNTF.

14 Perturbing hypotheses, connexions and data

Experts propose a hypothesis; the data either back it or not. To see what the data *can* back, we perturb, and there are three different things to perturb. They behave differently, so we keep them

apart.

- *The connexions* (a hypothesis about how copies are coupled). Contextuality-by-Default assumes the two copies of a variable disagree with probability exactly s_q, the least their marginals allow. A budget $\kappa_q \geq 0$ weakens this to "at most $s_q + \kappa_q$". The tables do not move.
- *The data.* Each window's table may be replaced by any table within total variation ρ of it (a data ball), and s_q is recomputed from the moved tables.
- *The hypothesis itself*, at level two: a latent-actor model whose fibre over the moving data deforms, and changes rung where it meets a wall.

All computations are in `engine/pce/ball.py`, checked by `engine/check_ball.py` (BL1–BL7).

The connexion budget (level one). Let $\lambda(\kappa)$ be the largest mass of a sub-measure on the copies whose window marginals lie below the tables and whose connexions disagree with probability at most $s_q + \kappa_q$. With no budget this is λ_C^{cbd}, the classical fraction of the consistified system (BL1: the disagreement form and the cellwise form agree).

Proposition 27 (budget profile)**.** *λ is nondecreasing, concave and piecewise linear in κ. Its one-sided derivatives (shadow prices) lie in $[0, 1]$. With a total budget $r = \sum_q \kappa_q$ allocated freely,*

$$\lambda(r) \;\leq\; \min\bigl(1,\; \lambda(0) + r\bigr).$$

Proof. κ enters only the right-hand sides of a linear program, so the value is concave and piecewise linear, and nondecreasing since the feasible set grows. For the bound, take an optimal sub-measure at budget κ. For each connexion, remove mass from atoms on which its copies disagree until its disagreement is at most s_q. This removes at most κ_q per connexion, hence at most r in all, and keeps every window constraint. The rest is feasible with no budget, so $\lambda(0) \geq \lambda(r) - r$. A unit of budget therefore never buys more than a unit of mass. □

On cyclic systems the bound is attained: $\lambda(r) = \min(1, 1 - c + r)$ on Snow Queen and on 12 random cyclic systems (BL2). So *the CbD degree c is the radius of the smallest connexion ball that makes the system noncontextual*, and any single connexion can carry the budget: the grading is flat, because the contextuality belongs to the cycle, not to a connexion. Grading becomes informative when a connexion lies on several cycles. Take two settings A_1, A_2 against three settings B_1, B_2, B_3, with one odd context (A_2, B_3) and small direct influences. The contextual cycles are the two through B_3. Relaxing the B_3 connexion alone reproduces the whole free profile, from $\lambda(0) = 0.73$ to 1 at $r = 0.28$; every other connexion relaxed alone stalls at 0.76 or below (BL3, BL4; fig. 10(b)). The shadow price of the total budget is 1 until saturation and 0 after.

Figure 10(c) shows the geometry. Project the polytope of all couplings of the copies (window tables exact) onto the agreement of the Kind/Evil copies and the summed agreement of the other three connexions. The CbD point (every connexion maximal) lies outside the polytope, and its distance to the best total agreement is exactly c.

The data ball (level one). Moving the tables changes both the window constraints and the allowances s_q. Because $s_q = |p - q| \geq \sigma(p - q)$ for $\sigma = \pm 1$, with equality for the right sign, the best support within the ball, $\max_{t'} \lambda(t')$, is exact as the maximum of one linear program per orientation; so is the smallest CbD degree within the ball, $c_{\min}(\rho)$. On cyclic systems the best

support is $1 - c_{\min}(\rho)$ (BL5). For Snow Queen (all responses), $c_{\min}$ falls linearly, by 8ρ on our grid, and reaches zero at $\rho^* = 0.017$ per window.

This must be read against sampling. Under multinomial resampling at the observed sizes, the median total-variation move of a window is 0.020 and its 95% quantile 0.044. So a coordinated worst-case move of every window, smaller than the typical random move of each, removes the contextuality (fig. 10(a,d)). The bootstrap (Study 8) still finds $c > 0$ in essentially every replicate, because random moves are not coordinated. Both statements are true. They answer different questions: "does resampling noise produce contextuality like this?" (no) against "is there a non-contextual system indistinguishable, window by window, from these data?" (yes). A claim of contextuality that is robust in the second sense needs ρ^* above the sampling scale, and these data do not have it.

The fibre (level two). For the two-type model on three binary variables, the fibre over a table is given in closed form (section 7). Along a path of tables leaving the model, the fibre moves continuously until it meets a wall (BL6). Every rung change is a sign change of Δ (weights leave $[0, 1]$: the signed rung), of Det (a complex-conjugate pair: the complex rung), or of a box coordinate (a susceptibility leaves $[0, 1]$). At the Det wall the real fibre runs off to infinity and returns as a complex pair with real part $\frac{1}{2}$ (fig. 10(e)). For a genuine mixture, the distance to the nearest wall is its *robustness radius*: every table within it keeps a genuine two-actor explanation, and beyond it the share of tables that do falls off, first through the box boundary (BL7; fig. 10(f), radius 0.031 for the example). The level-two examples are synthetic: Snow Queen has no three-way tables, and pairwise data do not determine one.

What is open. The quantum version of the budget (the connexion constraint is linear in the moment matrix, so it fits the relaxation of section 13); a joint, calibrated data ball (a simultaneous confidence region rather than a per-window radius); a level-two model written on the copies, so that the connexion budget and the latent fibre meet in one picture; and the exact form of $c_{\min}(\rho)$ (linear with slope 8 on our grid for this rank-4 system, not proved).

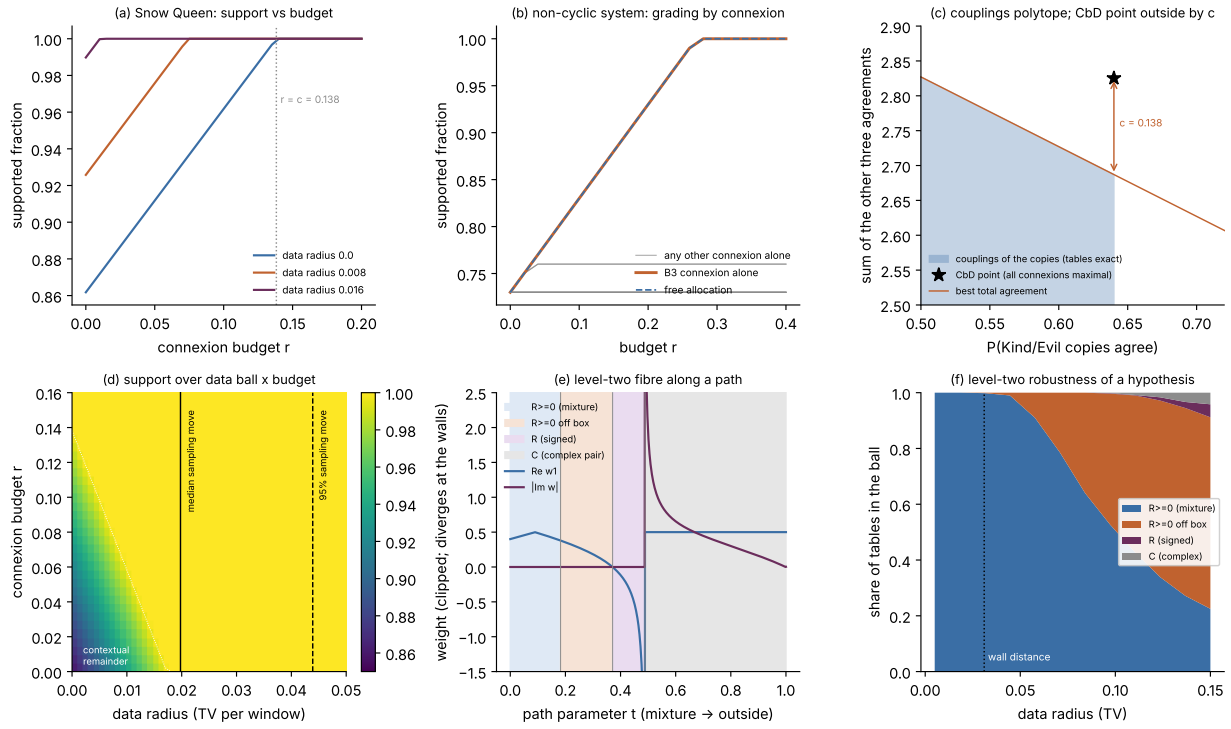


Figure 10: Perturbations. (a) Snow Queen (all responses): supported fraction against the connexion budget, for three data radii; it saturates at $r = c$. (b) A non-cyclic system: only the connexion on both contextual cycles (B_3) buys support. (c) The polytope of couplings projected on agreements; the CbD point lies outside by c. (d) Support over the data ball and the budget, with the median and 95% sampling moves of a window. (e) The level-two fibre along a path out of the model: walls where Δ, the box, and Det change sign; the weights diverge at the Det wall and return as a complex pair. (f) The share of tables in a data ball that keep each rung; the dotted line is the wall distance.

15 Contextuality and formal smoothness of the observable algebra

It is tempting to identify three kinds of regularity: reducedness of the Stanley–Reisner fibre at a merge point, vanishing of representation homology, and the equality conditions of the Islam–Volčič–Woerdeman positivity domains. They do not line up.

- A Stanley–Reisner ring is always reduced, and the merge point is a singular point of a reduced scheme. So reducedness is not smoothness.
- On a commutative fibre, derived representations with 1×1 coefficients give the Koszul homology of the moment equations. Its higher terms vanish exactly off the non-flat locus of section 7.
- Non-matrix coefficients only probe jets of a commutative algebra: $\mathrm{Hom}(A, k[\varepsilon]/\varepsilon^2)$ is the tangent space, and extensions between distinct points vanish.
- The hereditary quadratics belong to the windows of the quantum rung, not to the latent fibre.

Representation theory with arbitrary coefficients [66] bites where the algebra is noncommutative: the algebra of observables.

The observable algebra. Let Γ be a compatibility graph: its vertices are ± 1 observables, and its edges join commuting pairs. The observables generate the right-angled Coxeter group W_Γ, with $x_v^2 = 1$, and $x_u x_v = x_v x_u$ on edges. These are the relations used by the consistified relaxation of section 13. Quantum models in dimension n are points of

$$\mathrm{Rep}_n(W_\Gamma) = \{(X_v) \in M_n^V : \ X_v^2 = I, \ X_u X_v = X_v X_u \text{ on edges}\}.$$

Proposition 28. *For a finite graph Γ the following are equivalent:*

(i) *Γ is chordal: every cycle of length at least 4 has a chord.*

(ii) *No state and no measurements realising Γ produce contextuality [78].*

(iii) *W_Γ is virtually free.*

(iv) *$\mathbb{Q}[W_\Gamma]$ is formally smooth (quasi-free) in the sense of Cuntz and Quillen.*

Proof. (i)⇔(ii) is the theorem of Xu and Cabello. It rests on Vorob'ev's theorem: tables on the cliques of a chordal graph always glue.

(i)⇒(iii): a chordal graph is either a clique or splits along a clique separator. Accordingly W_Γ is a finite group $(\mathbb{Z}/2)^k$, or an amalgam over a finite clique subgroup. By induction it is the fundamental group of a finite graph of finite groups, hence virtually free.

(iii)⇒(i): an induced cycle C_k with $k \geq 4$ spans a special subgroup W_{C_k}. For $k = 4$ it is $D_\infty \times D_\infty$, which contains $\mathbb{Z}^2$. For $k \geq 5$ it is the reflection group of a right-angled hyperbolic k-gon, which contains surface groups. Neither occurs in a virtually free group.

(iii)⇔(iv): the Hochschild dimension of a group algebra is the cohomological dimension of the group over the field. Formal smoothness means Hochschild dimension at most 1 [79]. Over $\mathbb{Q}$, cohomological dimension at most 1 characterises the virtually free finitely generated groups [80]. □

Formal smoothness makes every Rep_n smooth. The converse is the falsifiable question: is the non-smoothness of a non-chordal Γ already visible in some matrix representation scheme, or does it need the non-matrix coefficients of [66]?

The test. At a point X we compute three things. The first is the Zariski tangent space Z^1, the kernel of the linearised equations. The second is the orbit dimension $n^2 - \dim$ of the commutant. The third is a second-order test. A tangent vector Y extends to second order if $L(Z) = -Q(Y)$ is solvable, where $Q(Y) = (Y_v^2; [Y_u, Y_v])$. If some tangent vector does not extend, X is a singular point, and with integer points and rational Y this is an exact certificate. We scanned every direct sum of characters (X_v diagonal ± 1, up to reordering), the most reducible points, where singularities concentrate.

- *Chordal graphs* (path, star, paw, diamond): no obstructed point in Rep_2 or Rep_3 (CX1), as proposition 28 requires.
- *Chordless cycles* C_4, C_5, C_6: no obstructed diagonal point in Rep_2, but obstructed points in Rep_3: 16 of 816, 160 of 5984, and 1024 of 45760 (CX2). Exact certificates exist for all three. At a certified point one rational tangent vector extends and another does not, so the extendable vectors form a cone, not a space (CX2, CX6).
- *The mechanism*, for $C_4 = K_{2,2}$, where $W = D_\infty \times D_\infty$. By the Künneth formula, H^1 of a character comes from one factor at a time, and $H^2 = H^1 \otimes H^1$. A point $\chi_i \oplus \chi_j \oplus \chi_k$ is obstructed exactly when $\chi_i\chi_k$ has H^1 from the first factor, $\chi_k\chi_j$ has H^1 from the second, and their cup product lands in $H^2(\chi_i\chi_j) \neq 0$. This predicts the obstructed set exactly, at all 816 points. The dimension $\dim H^1(W, \mathrm{End}\, V) = \dim Z^1 - \dim(\text{orbit})$ matches everywhere (CX3). It also explains why $n = 3$ is the first size among direct sums of characters: the cup product needs three distinct characters.
- *All graphs on* 4 *and* 5 *vertices* (45 isomorphism classes, 8 non-chordal): Rep_3 has an obstructed point if and only if the graph is not chordal (CX4). On the quantum side the same split appears (CX5). On chordal graphs the NPA maximum of random correlator functionals equals the classical one, to 3×10^{-9}. On each non-chordal graph an induced chordless cycle reaches the quantum n-cycle bound: $k\cos(\pi/k)$ for even k, and $(3k\cos(\pi/k) - k)/(1 + \cos(\pi/k))$ for odd k [81].

Proposition 29. *If* Γ *is not chordal and has an induced cycle of length* 4, 5 *or* 6, *then* $\mathrm{Rep}_3(W_\Gamma)$ *is singular. If* Γ *is chordal, every* $\mathrm{Rep}_n(W_\Gamma)$ *is smooth.*

Proof. Sending the generators outside an induced cycle C to 1 is a homomorphism $W_\Gamma \to W_C$, which retracts the inclusion $W_C \subset W_\Gamma$. So $\mathrm{Rep}_n(W_C)$ is a retract of $\mathrm{Rep}_n(W_\Gamma)$. A retract of a formally smooth scheme is formally smooth, because the lifting property passes to retracts. Hence a singular point of $\mathrm{Rep}_3(W_C)$, certified exactly for C_4, C_5, C_6 (CX2, CX6), gives a singular point of $\mathrm{Rep}_3(W_\Gamma)$. The chordal case is proposition 28(iv). □

Where contextuality is born. The direct sums of characters scanned above are the classical models: all observables commute, so they are mixtures of deterministic assignments. A natural conjecture is that contextual models branch off the classical locus at its singular points. For the CHSH scenario C_4 this can be settled, because $W = D_\infty \times D_\infty$ and its Hermitian models are classified. Each splits into tensor products of irreducibles of the two factors. These are characters, and two-dimensional irreducibles in which the two reflections meet at an angle $\theta \in (0, \pi)$. A model can violate the CHSH inequality for some state only if it has a $2 \otimes 2$ block. Off such blocks one side acts by characters, and the tables are classical. As $\theta \to 0$ or π a $2 \otimes 2$ block degenerates to four characters $\{\alpha, \alpha'\} \times \{\beta, \beta'\}$ with $\alpha\alpha' = (-,-)$ on one side and $\beta\beta' = (-,-)$ on the other. Call a

classical point containing such a block a *birth point*: contextual models come arbitrarily close to it, and by the classification to no other classical point. The checks (`check_birth.py`, BR1–BR5) give the following.

- *Born only at singular points.* Rep_3 has no birth point, since a $2 \otimes 2$ block needs dimension 4, yet it has 16 singular points. Rep_4 has 3876 classical points. Four are birth points, one per limit type, and all four are singular. So are another 240 that are not birth points (BR1, BR2). In Rep_5 every classical point containing a birth block is still singular (BR5).
- *Only there.* Perturb each point at random and project back onto the Hermitian locus of Rep_4. About a third (37%) of the models near birth points violate CHSH. None of 750 near non-birth singular points does, and none of 750 near smooth points (BR3).
- *At second order.* Along the birth curve, with both reflection angles equal to t, the largest CHSH value is exactly $2\sqrt{1+\sin^2 t} = 2 + t^2 + O(t^4)$ (BR4). Contextuality switches on at the order of the obstruction. Its tangent direction tilts both sides at once, and only birth points carry such a direction into contextual models.

So, for CHSH, singularity is necessary for the birth of contextuality, but far from sufficient. The singular points of Rep_3 show this most clearly: there the cup product blocks exactly the simultaneous two-sided deformation that contextuality would need. The classification argument is specific to C_4.

The 5-cycle: born at a smooth point. The pentagon (the KCBS scenario) has no block classification, so we define the contextuality of a model directly. By the complete set of n-cycle inequalities [81], a Hermitian model is contextual for some state if and only if

$$\max_{\gamma} \ \lambda_{\max}\Big(\sum_i \gamma_i X_i X_{i+1}\Big) > n - 2,$$

the maximum running over sign patterns γ with an odd number of minus signs. This needs no classification and works for any cycle. On C_4 it agrees with the block criterion. On C_5 the witness already reaches the quantum bound in Rep_3: KCBS lives in a qutrit, whereas CHSH needs dimension 4 (BR6). Up to the dihedral symmetry, sign flips of the observables and reordering, the 5984 classical points of $\mathrm{Rep}_3(C_5)$ fall into 33 classes.

- Exactly one class, of 160 points, gives birth to contextual models. The local maximum of the witness within distance ε is 7.5×10^{-3}, 6.8×10^{-4} and 7.3×10^{-5} at $\varepsilon = 0.1, 0.03, 0.01$, about $0.7\,\varepsilon^2$.
- That class is *smooth*. Its tangent dimension, 10, equals the dimension at nearby points, and its tangent vectors extend to third order (BR7).
- The singular class, the 160 obstructed points of proposition 29, has tangent dimension 10 against 8 nearby, and gives no birth. The largest nearby witness is 10^{-9} (BR8).

In the 5-cycle, then, birth and singularity are disjoint in Rep_3, and "born only at singular points" is a property of CHSH, not a general law.

What degenerates. The question is therefore not *where* the classical locus is singular but *what* degenerates onto a classical point. We record the block structure of nearby Hermitian models: the eigenspaces of a random element of the commutant, so a single block means irreducible. The scans are `engine/scan_c5.py` for $n = 3, 4$ and the checks BR9–BR11.

- *Rep*$_3(C_5)$. Every model near the birth class is irreducible, and 99.5–100% of them are contextual. Every model near the singular class splits as $2 + 1$, and none is contextual (BR9). At the birth point the KCBS family, irreducible 3-dimensional models, collapses onto three commuting characters. The point is smooth because that family's closure is a single smooth component there. At the singular class two families of reducible models meet, and a 2-dimensional block on a 5-cycle can never be contextual. In CHSH the same description holds: the birth points are where the irreducible 4-dimensional family ($2 \otimes 2$) collapses. They happen to be singular because reducible families pass through the same points.

- *Necessary, not sufficient* (BR10). Every birth class, in Rep_3 and in Rep_4, has a block of size at least 3, the smallest contextual dimension for C_5, degenerating onto it. The converse fails. In Rep_3 six classes have irreducible models nearby and one gives birth. In Rep_4, 99 of 242 classes have a block of size at least 3 nearby and 29 give birth. Nor is irreducibility enough globally: in a random sample of irreducible 3-dimensional models of W_{C_5} most are noncontextual for every state. The CHSH fact that every irreducible block of the smallest contextual dimension is contextual ($2\sqrt{1 + \sin\theta_A \sin\theta_B} > 2$) does not generalise. Simple saturation counts do not separate the birth class either: how many of its characters reach $n - 2$ on a common odd inequality does not single it out.

- *Singularity is a separate matter* (BR11). In $\mathrm{Rep}_4(C_5)$ the 52360 classical points form 242 classes. Of these, 20 classes (4360 points) are singular and 29 (6160 points) give birth, and they share only 5 classes (840 points). Neither set contains the other.

- *Order.* In most birth classes contextuality switches on at second order. The local maximum of the witness within $\varepsilon = 0.1$ is $7.5 \times 10^{-3} \approx 0.75\,\varepsilon^2$ in the KCBS birth class of Rep_3, and similar in most birth classes of Rep_4. Four birth classes of Rep_4 are of higher order (below).

	CHSH (C_4), Rep_4	KCBS (C_5), Rep_3	KCBS (C_5), Rep_4
classical points (classes)	3876	5984 (33)	52360 (242)
singular classical points	244	160	4360
birth points	4	160	6160
birth $\subseteq$ singular?	yes	no (disjoint)	no (overlap 840)
block degenerating at birth	$2 \otimes 2$, irreducible	irreducible 3	4 or $3 + 1$
irreducible block always contextual?	yes	no	no
order of the violation	t^2	ε^2	ε^2 (25 classes), higher (4)
second-order invariant $\Lambda_2 > 0$	exactly at birth	exactly at birth	exactly at second-order birth

Table 8: Where contextuality is born, CHSH against KCBS. Birth requires a large enough block degenerating onto the classical point, is independent of singularity, and is detected at second order by Λ_2 (proposition 30).

So contextuality is a property of what degenerates, not of the singularity (table 8). A classical point gives birth only if a non-commuting block of at least the smallest contextual dimension collapses onto it. Whether such a family actually enters the contextual region is a finer condition. Irreducibility, block dimension and saturation counts all fail to capture it. A second-order invariant does.

The second-order invariant. At a classical point X_0 each odd-sign cycle operator $B_\gamma = \sum_i \gamma_i X_i X_{i+1}$ is diagonal, with top eigenvalue $n-2$ on the characters that saturate γ; let P project onto them. Take a tangent vector Y that extends to second order, with lift Y_2. The equations of Rep_n are quadratic, so $F(z_0 + tY) = tJY + t^2 F_2(Y)$ exactly, and Y_2 solves $JY_2 = -F_2(Y)$. At singular points Y is restricted to the cone of extendable vectors. Along $X(t) = X_0 + tY + t^2 Y_2$ we have $B_\gamma(t) = B_0 + tB_1 + t^2 B_2$, and degenerate perturbation theory gives the top eigenvalue

$$(n-2) + t\,\lambda_1 + t^2\lambda_2 + O(t^3), \qquad \lambda_1 = \lambda_{\max}(PB_1P), \qquad \lambda_2 = \lambda_{\max}\big(PB_2P + PB_1(n-2-B_0)^+ B_1P\big).$$

Let Λ_2 be the maximum of λ_2 over odd γ and unit extendable Y (`engine/pce/birth2.py`).

Proposition 30 (the second-order invariant; INV1–INV3)**.** *(a) The first-order term vanishes: $\lambda_1 = 0$ at every classical point of CHSH (Rep_4, 67 classes) and of KCBS (Rep_3, 33; Rep_4, 242). Its diagonal part vanishes identically, because the involution equations force Y_v to have zero diagonal at a diagonal point. The off-diagonal part is computed, not proved.*

(b) $\Lambda_2 > 0$ exactly at the birth classes for CHSH ($\Lambda_2 = \frac{1}{2}$ at the one birth class) and for KCBS in Rep_3 ($\Lambda_2 = \frac{3}{4}$, matching the observed growth $0.747\,\varepsilon^2$). It is 0 at every other class.

(c) In $\mathrm{Rep}_4(C_5)$, $\Lambda_2 > 0$ at 25 classes, all of them birth classes, and $\Lambda_2 = 0$ at all 213 classes without birth. The remaining 4 birth classes have $\Lambda_2 = 0$ and are of higher order. There the largest witness of sampled nearby models scales like d^p with fitted p between 2.8 and 4.1.

The value $\frac{1}{2}$ for CHSH is consistent with the exact curve. Tilting the two reflections on each side symmetrically by $\pm\theta/2$ gives $2\sqrt{1+\sin\theta_A \sin\theta_B} = 2 + \theta_A\theta_B + \ldots$, where $\theta_A^2 + \theta_B^2$ is the squared length of that tangent vector in our coordinates; the growth is at most half of it, with equality at $\theta_A = \theta_B$. The KCBS values in Rep_4 are $\frac{3}{4}$, $0.433 \approx \sqrt{3}/4$, $\frac{1}{3}$, 0.30 and 0.26.

So the invariant that the block criteria missed is second-order and spectral. Contextuality is born at second order exactly where a tangent direction of the degenerating family makes the effective operator $PB_2P + PB_1(n-2-B_0)^+B_1P$ exceed $n-2$. When every such direction gives 0, it can still be born at higher order. In $\mathrm{Rep}_4(C_5)$ that happens at four classes. A higher-order invariant (third and fourth order in the same expansion) and a proof of (a) are the open steps. The universal fact is not quadratic growth but the vanishing of the first-order term.

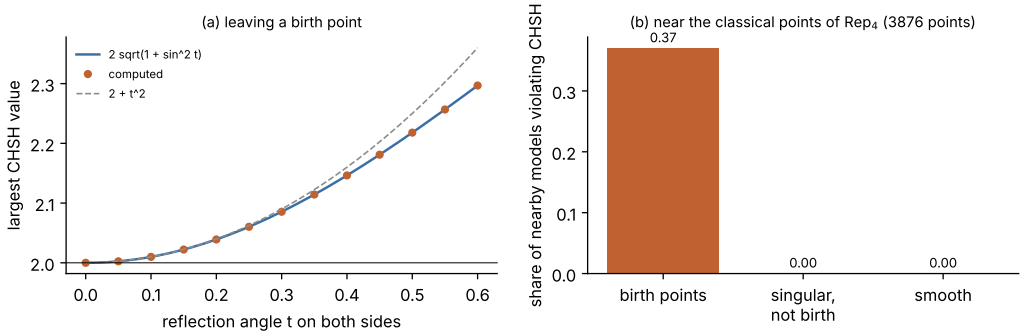


Figure 11: The birth of contextuality in the CHSH scenario. (a) Along the curve that tilts both pairs of reflections by t, the largest CHSH value is $2\sqrt{1+\sin^2 t}$: the violation grows like t^2. (b) Random models near each kind of classical point of Rep_4, projected onto the Hermitian locus: contextual ones occur only near birth points.

Reading. For these graphs, matrix coefficients suffice: 3×3 matrices detect the non-smoothness that group theory guarantees, and no non-matrix coefficient algebra is needed. The obstruction has a clear origin. A chordless cycle carries first-order deformations from its two "sides", and their cup product is the second-order obstruction. That same cycle is what lets quantum models be contextual, and for C_4 contextual models branch off the classical locus only at (some of) its singular points. For C_5 they branch off at a smooth point: the singular locus records where contextuality is possible, and the block that degenerates records where it is born. So contextuality and the singularity of the quantum representation schemes appear together, graph by graph (fig. 12).

Six limits apply. Birth is detected operationally, from 30 random nearby models and a local maximisation of the witness per class, so a very thin contextual region near a class could be missed. Only direct sums of characters were scanned, so smoothness of Rep_2 for the cycles is established only at those points. Cycles longer than 6 are not certified. Their singularity at $n = 3$ is a conjecture supported by the pattern. The second-order test certifies singular points, while the absence of obstructions on chordal graphs is a theorem, not a computation. We have not computed the full representation homology, only its first obstruction. And the link to contextuality is through the graph, not through a numerical identity between the size of the singular locus and any contextual fraction.

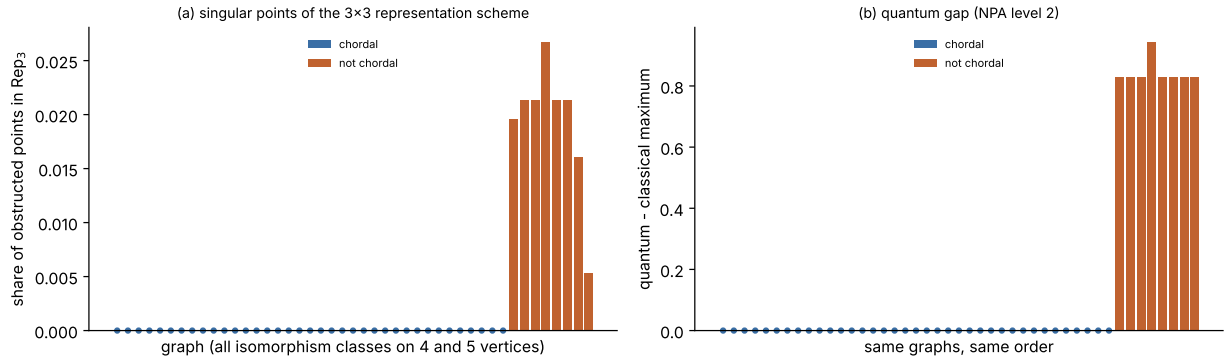


Figure 12: Every graph on 4 and 5 vertices, chordal (blue) and non-chordal (orange). (a) The share of direct sums of characters in $\mathrm{Rep}_3(W_\Gamma)$ with an obstructed tangent vector; it is non-zero exactly for the non-chordal graphs. (b) Quantum minus classical maximum (NPA level 2): zero on chordal graphs; on non-chordal graphs, the bound of the induced chordless cycle.

16 One grid: the posterior tower and the ladder

The work so far uses two towers. The *posterior tower* [82, 83], $X \times \mathcal{D}X \times \mathcal{D}^2X \times \mathcal{D}^3X \times \mathrm{Sh}X$, is a tower of depth: a predictive, a law over predictives, a law over those laws, and a shadow of labelled hypotheses. The *ladder* $\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\geq 0}$ of section 8 is a tower of coefficients: the number system the weights live in. They are two axes of one grid, and every result of this note and of [83] has an address in it.

For a commutative semiring S, let $\mathcal{D}_S X$ be the finitely supported S-valued weights on X that sum to the unit of S. It is a monad, with unit the Dirac weight and multiplication the barycentre. Taking $S = \mathbb{R}_{\geq 0}$ gives the distributions, $S = \mathbb{R}$ signed and $S = \mathbb{C}$ complex quasi-distributions, and $S = \mathbb{T} = (\mathbb{R} \cup \{\infty\}, \min, +)$ the tropical ones of [83, §11]. The tower, its context functor, the shadow $\mathrm{Sh} = \mathcal{D}(\Theta \times \mathcal{D})$, and the two candidate laws (shift and averaging) are all built from $\mathcal{D}$

alone, so each row S has its own tower $\mathcal{C}_3^S$.

Proposition 31 (the grid). *(a) A semiring map $\varphi\colon S \to S'$ induces a monad morphism $\mathcal{D}_\varphi$. Applied levelwise, it carries coherent contexts ($\mu\Pi = \rho$, $\mu\Omega = \Pi$, $\mathcal{D}(q)\Sigma = \Pi$) to coherent contexts. It commutes with both laws: $\mathcal{C}_3(\mathcal{D}_\varphi) \circ \lambda^S = \lambda^{S'} \circ \mathcal{D}_\varphi$. In particular this holds along $\mathbb{R}_{\geq 0} \to \mathbb{R} \to \mathbb{C}$. The tropical row is reached by the valuation, a monad morphism from the positive Puiseux row, not from $\mathbb{R}_{\geq 0}$ [83, Thm. 11.3].*

(b) The Beck table is the same in every row. $\mathcal{C}_3^S$ is a comonad. The averaging law satisfies the unit, counit and multiplication axioms and fails comultiplication. The shift law satisfies the counit axiom and fails unit and multiplication.

(c) What needs positivity lives only in the $\mathbb{R}_{\geq 0}$ row: convex order and Strassen's theorem, the martingale reading of a coherent context [83, Thm. 5.3], pooling as conditional expectation, data processing, and measuring a Beck defect by a convex-order diameter. Over $\mathbb{R}$ a coherent level-two law can violate Jensen's inequality. The signed barycentre fibre is unbounded, so a Beck defect has no size there.

Proof. (a) Everything is a finite sum of finite products of weights, and φ preserves both. (b) The verifications in [82] use only associativity, commutativity, distributivity and the unit. The counterexamples differ in which atoms carry which contexts, not in the values of the weights. With the weights $\frac{1}{2}, \frac{1}{2}$ replaced by any admissible pair (in $\mathbb{T}$, by $0, 0$), they remain counterexamples. (c) Let v be such that $\rho \pm v$ are distributions, and take $\Pi_t = (1+t)(\frac{1}{2}\delta_{\rho+v} + \frac{1}{2}\delta_{\rho-v}) - t\delta_\rho$. It has barycentre ρ for every t, and its second moment exceeds that of δ_ρ by $(1+t)\|v\|^2$. The mirrored family falls below it by $t\|v\|^2$, against Jensen. So the signed fibre is unbounded in both directions. □

All three parts are checked in `check_grid.py` (TG1–TG6), which re-implements the tower code of [82] with the semiring as a parameter. It runs the full Beck table in the four rows $\mathbb{R}_{\geq 0}, \mathbb{R}, \mathbb{C}$ (exact Gaussian rationals) and $\mathbb{T}$, and finds it identical (TG1–TG3). It checks change of base along $\mathbb{R}_{\geq 0} \to \mathbb{R} \to \mathbb{C}$ (TG4) and gives the signed counterexamples of (c) (TG5). TG6 shows the two row splits that this note is about. At level one, the PR box has a section in the $\mathbb{R}$ row and none in the $\mathbb{R}_{\geq 0}$ row (contextuality). At level two, a three-variable table has a complex two-type fibre and no real one.

Where each result lives. Depth is read off the tower: level one is a law on global outcomes (gluing), level two is a law over predictives, and so on. The row is the semiring in which the weights are allowed. Table 9 places the results; "G" refers to sections of [83].

The last row of table 9 is a third direction, not another semiring. Replacing commutative weights by positive semidefinite matrices does not give a monad on sets in the same way. Its classical points are the $\mathbb{R}_{\geq 0}$ row embedded as diagonal matrices. Section 15 shows that contextuality is born where that embedding sits singularly in the matrix representation scheme. Section 12 supplies the categorical frame for this row: modules with positivity, and NPA as truncated free modules.

row	level one: $\mathcal{D}_S$ (gluing)	level two: $\mathcal{D}_S^2$ (latent actors, fibres of μ)
$\mathbb{C}$	Kirkwood–Dirac sections (section 9)	complex fibres, Det < 0 (sections 7 and 8); monodromy (section 6)
$\mathbb{R}$	signed sections: no direct influence, linear gap 0; negativity (section 8)	signed mixtures, Det $> 0 >$ CbD (section 7)
$\mathbb{R}_{\geq 0}$	gluing, λ^*, graded slices (section 2); CbD and consistified fractions (section 13); budgets and data balls (section 14); decisions over time (section 17)	genuine mixtures; strata, Puiseux crossings, walls (sections 3 to 5 and 14); convex order, the martingale, Beck defects (G§§2–7)
$\mathbb{T}$	tropical tables (G§11)	tropical secants, Berkovich factorisation (G§§11–12)
matrices	quantum fraction λ_Q, NPA (sections 12 and 13); positivity domains (section 11)	the observable algebra and Rep_n; birth of contextuality (section 15)

Table 9: The grid: tower depth (columns) against coefficients (rows). Moving up a row relaxes positivity; contextuality and the complex rung are row splits at fixed depth. The last row is not a semiring: it replaces commutative weights by positive semidefinite matrices.

17 Deciding on a time-varying sheaf: covers, awareness and robust satisficing

The note so far takes the evidence as given and asks what it supports. A decision maker faces three further facts. The evidence arrives over time. New contexts add windows, and new distinctions refine the outcomes. Because the evidence does not glue, the decision maker must also choose *what to glue*. This section turns the level-one machinery into a decision procedure for that situation. It works one small example through completely (figs. 13 and 14). Everything is in `engine/pce/timesheaf.py`, `engine/timesheaf_example.py` and `engine/check_timesheaf.py` (TS1–TS8).

The objects. At time t the evidence is a presheaf of local laws on a context complex Δ_t, whose faces are the observed windows. A step $t \to t'$ is given by three things.

- An inclusion $\Delta_t \hookrightarrow \Delta_{t'}$: new windows appear.
- For each variable, a surjection r_v from its new outcomes onto its old ones. This is awareness growth: an outcome splits into finer ones.
- A *drift* d: the largest total-variation distance between an old table and the corresponding new table coarsened along r. The step has no drift when $d = 0$.

A *cover* C is a set of observed windows that the decision maker chooses to glue. For an act a with utility u_a and $\alpha \leq \lambda^*(C)$, the robust value is

$$V_C(a;\alpha) = \alpha \min_{q \in E_\alpha(C)} \mathbb{E}_q[u_a] + (1-\alpha) \min_x u_a(x).$$

The explained part of the world is a law q that accounts for a fraction α of every table in C. The unexplained part is treated as adversarial. This is Huber's ε-contamination with $\varepsilon = 1 - \alpha$ [84],

and E_α is a credal set in Walley's sense [85]. What is new is that the smallest admissible ε, namely $1-\lambda^*(C)$, is set by the data's own incoherence rather than chosen by hand. For a requirement u_0 ("good enough"), the robustness of (C, a) is the largest $\varepsilon \in [1-\lambda^*(C), 1]$ such that $V_C(a; 1-\varepsilon') \geq u_0$ for every ε' up to it. This is info-gap robustness [87]. Robust satisficing chooses the cover and act with the largest robustness [88, 89]. Each value is one linear program.

Proposition 32 (covers and time)**.** *(a) If $C \subseteq C'$ then $\lambda^*(C) \geq \lambda^*(C')$, and $E_\alpha(C') \subseteq E_\alpha(C)$ for $\alpha \leq \lambda^*(C')$. More evidence is less glueable but more informative (TS1).*

(b) An acyclic cover whose windows agree on their overlaps glues: $\lambda^* = 1$ (Vorob'ev [93]). With direct influence even a tree can fail (TS2).

(c) A step without drift never raises λ^*, and coarsening maps $E_\alpha(t')$ into $E_\alpha(t)$ (TS3, TS4). New windows and new distinctions can reveal incoherence but never remove it.

(d) A pendant window leaves λ^* unchanged when its marginal on the shared variable equals an existing window's. A pendant window is one whose other variables are new and which meets the old windows in one variable. With a different marginal it can lower λ^* (TS4).

(e) If every window moves by at most d, then $|\lambda^*(t') - \lambda^*(t)| \leq m\,d$, where m is the number of windows (TS5).

Proof. (a) Every constraint of C is among those of C'.

(b) Glue along the tree by conditional independence; the overlaps agree.

(c) A sub-measure feasible after the step, coarsened along r and restricted to the old windows, stays below every old table. These are the old tables, because there is no drift.

(d) Extend an old explanation μ by the new window's conditional law given the shared variable v. The extension fits under the new table iff $\mu_v \leq t'_v$. If t'_v is an existing window's marginal on v, this already holds.

(e) Remove from an optimal sub-measure the atoms that exceed a moved table. Each window loses at most d. □

The worked example. The truth is that of example 3: two patient types and two populations. Three acts are available: the triple combination $A{+}B{+}C$, the pair $A{+}B$, and nothing. The utilities are benefit minus harm. The triple has $1 - 1.2\,[\text{any harm}] - 0.5\,[\text{severe } C \text{ harm}]$, the pair has $0.75 - 1.2\,[\text{harm from } A \text{ or } B]$, and nothing has 0. The requirement is $u_0 = 0.05$. An expert also proposes the hypothesis "the triple's harm is at most 0.60". Its *support* is the largest α for which E_α contains a law satisfying it. Five times:

time	what happens	λ^*	support of the hypothesis	robust choice (robustness)	reading
t_0	AB, AC trials (population 1)	1	1	triple on $AB{+}AC$ (0.09)	a tree glues
t_1	+ BC trial (population 2)	0.860	0.860	triple on $AB{+}AC{+}BC$ (0.20)	conflicting evidence still narrows the harm
t_2	C harm split into mild and severe	0.860	0.849	pair on AB (0.04)	expires: a new distinction enters the utility
t_3	AB drifts ($d = 0.047$)	0.874	0.847	pair on AB (0.04)	within $m\,d$; the tree $AB{+}AC$ now fails (0.953)
t_4	+ biomarker window BD	0.874	0.847	pair on AB (0.04)	pendant: nothing changes

Table 10: The worked example. The hypothesis is "triple harm ≤ 0.60".

Figure 13 shows each time twice. Above is the context complex: the windows are the faces, each carrying its table, and the cycle A–B–C is shaded once it closes. Below is the decision plane, with the expected utility of the triple against that of the pair. The explanation sets E_{λ^*} and $E_{0.75}$ project to polygons. The hypothesis cuts $E_{0.75}$ (hatched). The background shows which act is best where, and the star marks the robust values. Three features are worth reading off.

- *Conflict can help.* At t_1 the BC trial comes from another population and the tables stop gluing, yet the robustness of the triple rises from 0.09 to 0.20. The trial still constrains C, and the narrower polytope outweighs the larger unexplained part. Satisficing on the full cover beats discarding the inconvenient trial.

- *Awareness growth can end a decision that every evidence radius protects.* At t_2 nothing about the evidence changes (λ^* stays 0.86, the drift is 0), but the severe outcome can now be told apart. It carries its own cost, so the triple stops being good enough. Robustness radii protect against changes in evidence, not against new distinctions entering the utility. That is a limit of the procedure, and the example shows it (fig. 14(c)).

- *The hypothesis loses support before the evidence does.* The support of "triple harm ≤ 0.60" equals λ^* until t_1. After the refinement and the drift it falls below λ^* (0.849 against 0.860, then 0.847 against 0.874). The hypothesis cuts off part of the best-supported explanations, so it now asks the data to give up more than their incoherence forces.

Figure 14 follows the covers. Panel (a) gives λ^* of every sub-cover over time, the full cover and the hypothesis's support. Panel (b) gives the frontier of covers in (glueability, informativeness) at t_1 and t_3: what the decision maker can choose to glue, and what each choice costs. Panel (c) gives the robustness of each act on its best cover and on all windows, with the decisions taken.

The selection hazard. Choosing the cover after seeing the data is a selection step, and it flatters (TS7). On 40 structureless random systems, the best proper sub-cover glues 0.26 more than the full cover on average. In 7 of them, no act is good enough on all the evidence, yet one becomes "good enough" on a sub-cover chosen afterwards. The procedure therefore has a rule, not a footnote: report the whole frontier, or fix the cover-selection rule before the data are seen and freeze it, as for the studies of section 13.

Place in the literature. The decision rule is standard. Robust satisficing [88, 89] and info-gap [87] choose the option that stays good enough over the widest set of circumstances. Maxmin expected utility [86] and credal sets [85] supply the sets. Satisficing under uncertainty, as opposed to under risk, is the regime where it is defensible [90]. Choosing what to keep from inconsistent evidence is the repair problem of inconsistent databases [91]. Growing awareness has its own decision theory [92]. What this section adds is the following:

- the ambiguity set is derived from the data as the graded explanation sets;

- its smallest contamination is fixed by λ^*;

- covers are compared on an exact frontier;

- the time-varying sheaf supplies the monotonicity and Lipschitz results of proposition 32.

Limits. The example is synthetic and small, and the decisions use level one only. The fibre and its walls (section 14) would add structural events, but they are not exercised here. Robustness is measured in contamination alone. The connexion budget and the data ball of section 14 would add two further radii. The expiry at t_2 is not predicted by any radius; it is visible only by recomputing.

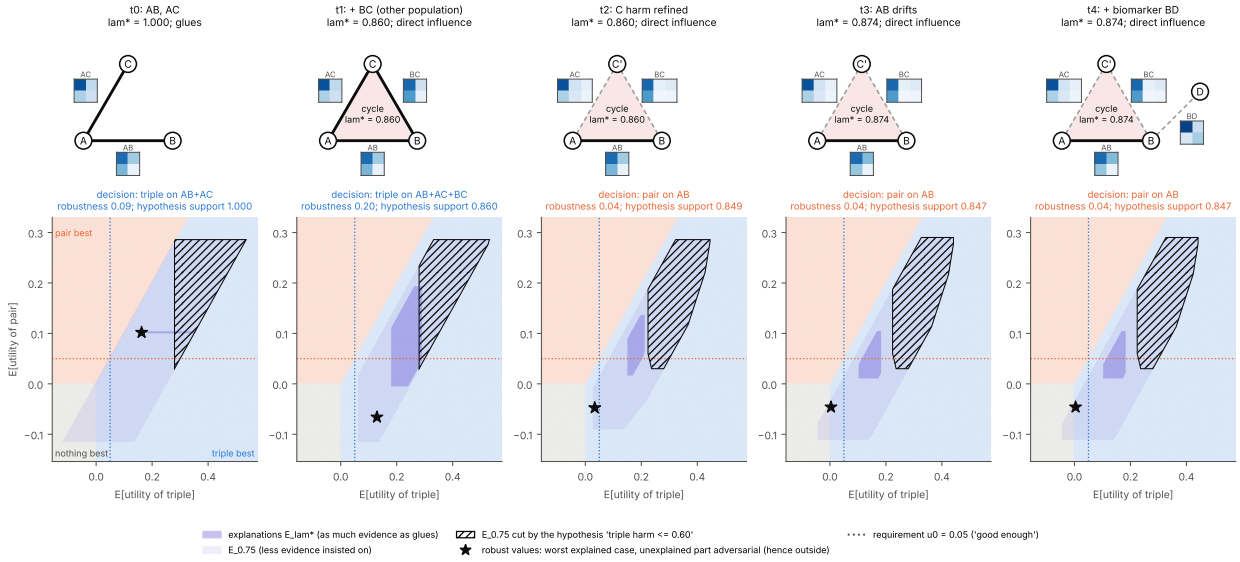


Figure 13: The worked example over five times. Top: the context complex, with each window drawn as a face carrying its table. The cycle is shaded red when it fails to glue, and the windows of the chosen cover are solid. C' marks the refined outcome of C. Bottom: the decision plane, with expected utility of the triple against the pair. The polygons are the explanation sets E_{λ^*} (dark) and $E_{0.75}$ (light); the hatched part of $E_{0.75}$ satisfies the hypothesis "triple harm ≤ 0.60". The tint shows the best act, and the dotted lines the requirement. The star marks the robust values, which lie outside because the unexplained part is treated as adversarial.

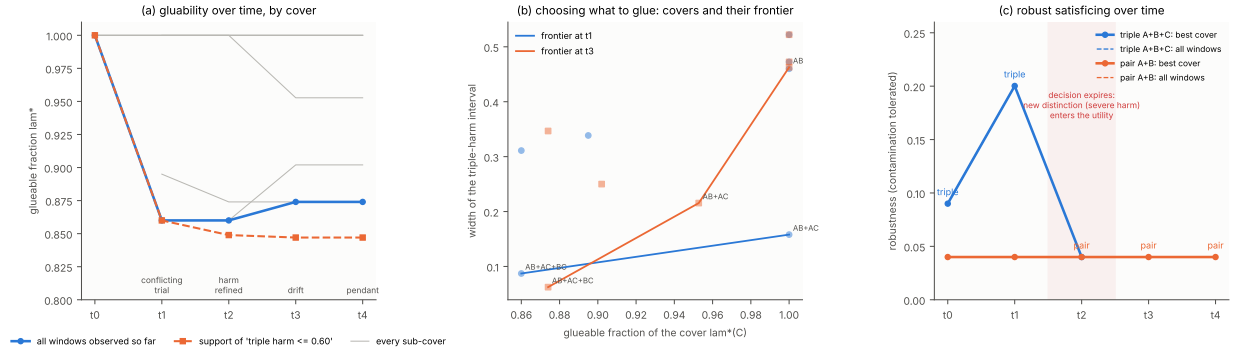


Figure 14: (a) Glueable fraction over time: every sub-cover (grey), all windows observed so far (blue), and the support of the hypothesis (dashed). (b) Covers at t_1 and t_3 by glueable fraction and width of the triple-harm interval, with their frontiers. (c) Robustness of each act on its best cover (solid) and on all windows (dashed), with the decision taken at each time. The decision expires at t_2, when the severe-harm distinction enters the utility.

18 What this says about probability built on schemes

Table 11 collects the dictionary. Level one is linear: explanations form polytopes, the gradation α slices them, and everything is a linear program. The only way the structure changes is through the combinatorics of the slices. Level two is polynomial: explanations are the real points of the fibres of F, the fibres have singular points where the actors merge or vanish, the obstruction to deforming an explanation is quadratic, and closed loops of evidence can permute the actors. The gradation lives at both levels (λ^* and λ_2^*), and so a programme for a theory of probability on schemes is: for each level, the fibre over the evidence; for each fibre, its graded slices; and for each path of evidence, the path it induces through the slices, including where it must pass through a singularity and what is lost or born there.

Related work. The noncontextual fraction and the sheaf view of local tables are due to Abramsky and coauthors [1, 3]; graded agreement in sheaves of pseudometric spaces is Robinson's consistency radius [11]; the companion report [31] develops the level-one engine. That latent-class and other hidden-variable models are stratified, singular algebraic families is classical in algebraic statistics [8, 9], and the consequences of their singularities for estimation and testing are studied by Drton [7] and Watanabe [30]. Generic identifiability of two-type models on three or more binary variables is due to Allman, Matias and Rhodes [4]. The secant-cumulant coordinates of section 7 and the toric structure of secant varieties are due to Michałek, Oeding and Zwiernik [24]; the semialgebraic description of tensors of nonnegative rank two to Allman, Rhodes, Sturmfels and Zwiernik [12], with a book-length treatment of semialgebraic latent models by Zwiernik [26]; real rank and the hyperdeterminant to de Silva and Lim [16] and Gelfand, Kapranov and Zelevinsky [17]; Stanley–Reisner theory and flatness are standard [25, 23]. What we add is the combination: graded partial gluing at the failure of a sheaf condition, lifted to the level of hypotheses, with the singular strata, second-order obstruction, Puiseux crossing and monodromy read as what happens to latent actors when evidence moves.

Limits. The categorical section is a framework, not a theorem: its computations (NPA, the Peres–Mermin square, the $q^n = 1$ obstruction, data processing) are standard or elementary, and the monotonicity of the quantum fraction under processing is checked, not proved. The quantum section uses three qubits with pairwise windows; pure fractions are lower bounds from a non-smooth optimisation; topology changes of the normalised quantum slices were not established, because their maximisers come in continuous families. The masking study uses two qubits with random states and directions, six trials per level, and a Gaussian χ^2 calibration that proved somewhat optimistic; the positivity-only search samples the feasible set by random walks and does not certify its extent. The non-commutative section uses spin models and small random examples; its molecular reading is an analogy, not chemistry, and its scheme-theoretic part (reducedness with coherences, the Rep_d tower) is stated, not computed. The exact scheme results are for two types: the Stanley–Reisner fibre for $m = 3, 4$, the closed form and the stratification for $m = 3$, and one flat point for $m = 4$; the level-two ladder is for $m = 3$, and the level-one comparability constants are measured, not derived, beyond the upper bound. The fibres are semialgebraic (they carry the inequalities $w \geq 0$, $0 \leq a \leq 1$, $\alpha F \leq t$), so "scheme" is used for the equations and the real points with these inequalities are the explanations. All variables are binary, and the level-two results are for two types. The values of λ_2^* and the level-two intervals come from multi-start optimisation and are certified only as attained (lower bounds for λ_2^*, inner bounds for intervals); the regularity of the 4-variable loop is checked at 500 points, not proved. Apart from Study 8 (section 13), the examples are synthetic with exact tables, with no sampling error. Study 8 is a single published experiment whose conclusion

probabilistic notion	geometric form
the tables glue	$\lambda^* = 1$: the level-one fibre (a polytope) is non-empty
graded partial gluing	the slices E_α of the polytope $\{\mu \geq 0 : \mu\|_w \leq t_w\}$, $0 < \alpha \leq \lambda^*$
dissipated mass	the residues r_w of (1); $1 - \alpha$ of every table
most probable path	the entropy maximisers q_α, continuous in α
latent actors	the fibre $\mathcal{X}_P$ of the polynomial map F, an affine scheme
actors merge, or one is absent	singular strata; over an independent law, the Stanley–Reisner scheme of a simplicial complex
actors partly invisible	the non-flat locus $\{w_1 w_2 d_1 d_2 d_3 = 0\}$ $(m = 3)$
second-order obstruction ($\ell_2 \neq 0$)	$\kappa = C^\top D^2 F$; tangent cone = union of branches
abrupt change	a path of laws crossing a fold of the image; Puiseux branches $\pm\sqrt{s}$
a structure dies, a new one is born	real points of the fibre become a complex-conjugate pair (sign of the hyperdeterminant); three types fit
signed mixtures	real points with weights outside $[0, 1]$ $(\mathrm{Det} > 0 > \Delta_m)$
kinds of incoherence	rungs of the ladder $\mathbb{C} \supset \mathbb{R} \supset \mathbb{R}_{\geq 0}$: no points (direct influence), $\mathbb{R}$ not $\mathbb{R}_{\geq 0}$ (contextuality, negativity ν), $\mathbb{C}$ not $\mathbb{R}$ (level two)
commutation obstruction	the cover of contexts; order effects $\leq \|[P, Q]\|$; Kirkwood–Dirac section on the $\mathbb{C}$ rung
loops of evidence	geometric phase; non-abelian holonomy with rank-two projections
masked measurements	linear ambiguity of dimension $\max(0, 3(3 - \dim V) - h)$; the cone culls candidates implying negative probabilities; recoverable = invariants of the residual group
categorical home	contexts as a diagram of commutative subalgebras; contextuality as a failed limit; NPA truncation; $q^n = 1$ obstruction; data processing
origin of observations	nested model-class fractions with direct influence and CbD below them; consistified fractions split $1 - \lambda^*$ into direct influence ($\leq \sum_q s_q$), quantum-explainable and beyond-quantum contextuality
perturbing hypotheses	connexion budget: concave profile, price ≤ 1, c = radius of the smallest noncontextual ball; data ball: $c_{\min}(\rho)$; level two: wall distance as robustness radius
where contextuality is possible	non-chordal compatibility graph $\Leftrightarrow$ the observable algebra $\mathbb{Q}[W_\Gamma]$ is not formally smooth; singular points already in Rep_3 (cycles of length 4–6); contextual models are born where a large enough non-commuting block degenerates onto a classical point; detected at second order by the effective operator Λ_2; singular for CHSH, not in general
the two towers	one grid: tower depth against base semiring; Beck table the same in every row; order and martingale facts only in $\mathbb{R}_{\geq 0}$
deciding over time	a time-varying sheaf with covers chosen by robust satisficing; contamination fixed by λ^*; new windows and distinctions never raise λ^*; drift moves it by at most $m\,d$
quantum rung	positivity domain of $\bigoplus_w (\rho_w^\top - K_w^* K_w)$; walls in the free locus $\det f_w = 0$; $\lambda_Q \leq \lambda^*$ in every product basis
probability paths	Zeno dragging: success $\cos^{4jN}(\Theta/2N) \to 1$, only with non-commuting projections
dual roles	monodromy of F (a subgroup of S_k); trivial on real mixtures when $m = 3$ because the non-flat walls separate the sheets

Table 11: The dictionary.

was known in advance, analysed from aggregated proportions; the quantum fractions with direct influence are computed by a relaxation that is not strictly feasible (accuracy about 2×10^{-4}), whose registered level $(1{+}E)$ is not tight.

Next steps. (1) Certified level-two bounds by polynomial optimisation (moment and sum-of-squares relaxations), which would give upper bounds on λ_2^* to match the lower ones. (2) The images of the three-type model and their folds: where the third actor of section 5 is itself forced to merge or vanish. (3) Monodromy for $k \geq 3$ (subgroups of S_k) and its relation to the singular strata. (4) A pre-registered test on aggregated combination-trial tables, if published pair and triple tables can be found; nothing patient-level would be needed. (5) Schemes that see positivity: over $\mathbb{R}$ or $\mathbb{C}$ the linear equations of level one always have solutions when the tables agree on overlaps, so contextuality is invisible to ordinary schemes; it is a statement about $\mathbb{R}_{\geq 0}$-points. Semiring schemes (blueprints [22], tropical schemes [18]) are the natural home for gluing with positivity, with λ^* and the graded explanations as their points over the α-line. (6) The flattening stratification for $m \geq 4$ and $k \geq 3$. (7) Non-commutative level two: whether the fibre over a merge point stays reduced once coherences are allowed (qubit actors), the quantum rung by the NPA hierarchy alongside λ^*, and whether laws on the complex rung of section 8 have qubit-actor explanations. (8) A pre-registered order-effect test (the QQ equality and the commutator bound) on aggregated survey data. (9) The quantum slices: topology of the normalised explanation sets as the evidence moves (the analogue of Example 4.3 of [54]), with continuous symmetries quotiented out; and the meaning of the higher levels of the free domain. (10) A scenario where the module level genuinely exceeds every fixed dimension (multi-outcome or multipartite, or a Slofstra-type game), and representation homology with non-matrix coefficients [66] as a test of where finite explanations become singular.

Reproducibility. The engine module is `engine/pce/level2.py`; `engine/check_level2.py` runs nine checks (LV1–LV9: graded explanations, the conflicting trials, the exact second derivative, the strata, the tangent cone, the Puiseux crossing, monodromy, sharpening, and level-two graded gluing) and writes `level2/level2_results.json`; `level2/fig_level2.py` draws fig. 1. `engine/check_scheme.py` runs six exact symbolic checks (SC1–SC6: the moment normal form, the Stanley–Reisner fibre and its primary decomposition, the closed-form inverse and Jacobian, the hyperdeterminant, the $m = 3$ strata, and the flat point for $m = 4$) in a few seconds and writes `level2/scheme_results.json`; `level2/fig_scheme.py` draws fig. 2. The exploratory scripts are in `level2/scheme/`. `engine/pce/ladder.py` computes the rungs and their measures; `engine/check_ladder.py` runs six checks (LD1–LD6) and writes `level2/ladder_results.json`; `level2/fig_ladder.py` draws fig. 3. `engine/pce/noncomm.py` implements spin rotations, sequential measurement, the commutation cover, the Kirkwood–Dirac section, Bargmann invariants and Zeno dragging; `engine/check_noncomm.py` runs eight checks (NC1–NC8) and writes `level2/noncomm_results.json`; `level2/fig_noncomm.py` draws fig. 4. `engine/pce/masked.py` implements the two-qubit tables, the linear and positivity-constrained fits, the lattice culls, sweeps and the feasible-region search; `engine/check_masked.py` (MK1–MK6, about 13 minutes) writes `level2/masked_results.json`, `engine/check_masked_gauge.py` (MK7) checks the dimension formula, and `level2/fig_masked.py` draws fig. 5. `engine/pce/qglue.py` builds the hereditary quadratics, their linearisations and the pure and mixed quantum fractions (semidefinite programs solved with SCS through cvxpy); `engine/check_qglue.py` (QG1–QG7) writes `level2/qglue_results.json`; `level2/fig_qglue.py` draws fig. 6. `engine/pce/npa.py` implements the NPA levels 1, $1{+}AB$ and 2 for bipartite binary scenarios and the dimension-free quantum fraction; `engine/check_npa.py` (NP1–NP6) writes `level2/npa_results.json`;

`level2/fig_npa.py` draws fig. 7. `engine/pce/fingerprint.py` computes the origin fingerprint; `engine/check_fingerprint.py` runs FP1–FP7. `engine/pce/cbdfrac.py` computes the consistified fractions; `engine/check_cbdfrac.py` runs CF1–CF7 and writes `level2/cbdfrac_results.json`; `check_grid.py` at the package root (TG1–TG6) gives section 16. `engine/pce/timesheaf.py`, `engine/timesheaf_example.py` and `engine/check_timesheaf.py` (TS1–TS8, writing `level2/timesheaf_results.json`) give section 17; `level2/fig_timesheaf.py` draws figs. 13 and 14. `engine/pce/coxeter.py` and `engine/check_coxeter.py` (CX1–CX6, about ten minutes, writing `level2/coxeter_results.json`) give section 15, with `engine/check_birth.py` (BR1–BR11, using `engine/scan_c5.py` for the 5-cycle scans, writing `level2/birth_results.json`; `level2/fig_birth.py` draws fig. 11) and `engine/check_invariant.py` (INV1–INV3, using `engine/scan_invariant.py`, writing `level2/invariant_results.json`); `level2/fig_coxeter.py` draws fig. 12. `engine/pce/ball.py` and `engine/check_ball.py` (BL1–BL7, writing `level2/ball_results.json`) give section 14; `level2/fig_ball.py` draws fig. 10. Study 8b is `PREREG_study8b.md`, `analyze_study8b.py` and the post-hoc scripts in the same folder; `level2/fig_snowqueen2.py` draws fig. 9. Study 8 is in `engine/study8_snowqueen/`: the frozen pre-registration and analysis (`FROZEN_PREREG_study8.txt`), the transcribed aggregated tables, the deviations log and `results_study8.json`; `level2/fig_snowqueen.py` draws fig. 8.

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